A signal–noise phase theory of when dendritic structure helps local credit assignment

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Abstract

Credit assignment requires neural circuits to distribute output errors across synapses, yet it is unclear when a dendritic tree improves this distribution. We show that exact gradients in conductance-based neurons factor into synapse-local eligibility and a transported compartment error. A credit-operator bound predicts when restricted routes help from their retained task signal, bias, gain and noise; its utility predicted trained outcomes across independent route and depth experiments (Spearman $\rho = 0.937$ and 0.916). Per-neuron feedback recovered most of the loss from a scalar signal. At fixed budgets, depth alone reduced accuracy in every seed. Across two hierarchy depths of a mechanism-matched task, aligned serial divisive composition improved accuracy by approximately 31 percentage points while literal grouped-point controls stayed flat; placement reversal abolished or reversed the effect. Flexible parameter-matched point networks nevertheless performed about 7 points better. Reconstructed arbors from one MICrONS mouse supplied sparse, coarse ancestry addresses, capturing most dense-field energy with approximately 7% of the wiring but only an approximately 2.7-fold anatomy-specific efficiency at matched density. Focal shunting localized credit only in permissive high-conductance regimes, and measured visual responses showed no morphology-specific alignment. Thus neuronal signals provide coordinates, dendritic ancestry supplies addresses, conductance regulates route gain and task–morphology alignment determines utility.

Introduction

A biological neuron must distribute credit for its output errors across thousands of synapses arrayed on a branching dendritic tree, a problem that the point units of standard neural networks do not face [1–3]. Each synapse samples presynaptic drive, local voltage, reversal potential and input resistance. By increasing local conductance, inhibition can shunt excitation and alter the gain of every dendritic path through that compartment [4, 5].

Backpropagation applies the chain rule from an objective to every parameter [6]. It is therefore the exact optimization reference, not itself a biological mechanism: a literal implementation requires parameter-specific derivatives, coordinated backward Jacobians and suitable forward–feedback relationships. Synapses instead access presynaptic activity, local postsynaptic state and modulatory signals of uncertain spatial precision [7, 8]. Credit assignment in a neuron consequently has two spatial levels: identifying the neuron that should change and distributing that coordinate through its arbor. We therefore ask when a restricted signal can approximate the exact dendritic error field.

Biologically motivated approaches relax different requirements of backpropagation. Three-factor rules combine local eligibility with a modulator [9]; feedback alignment replaces transposed weights with alternative pathways, requiring positive functional alignment rather than exact symmetry [10, 11]; predictive, equilibrium and prospective schemes infer desired neural states [12–14]; and eligibility-propagation methods separate online synaptic traces from neuron-specific learning signals [15]. Dendritic-error and burst models separate forward and teaching signals through compartments, interneurons or temporal codes [16–20]. Most deliver a neuronal coordinate or use a few named compartments, leaving open what a full arbor contributes after credit reaches one neuron.

Our earlier regular-tree study derived the exact gradient required by a conductance-based dendritic network and tested its approximation from restricted somatic feedback [21]. Each gradient factorizes into *local eligibility* $\times$ *compartment error*: eligibility depends on presynaptic activity, driving force and input resistance, whereas a soma error is transported through path-specific gains. This identifies restricted feedback as a bottleneck but does not show whether real arbors provide useful addresses, whether conductance regulates them, or whether they align with task gradients.

Here we address those questions by formalizing a coordinate–address–gain hierarchy inside a common credit-operator theory. Teaching signals provide signed neuronal coordinates; dendritic ancestry provides nested candidate addresses; and conductance determines route gain. The theory predicts a phase boundary: a restricted route helps stochastic learning only when the useful gradient signal it retains compensates for its representation bias, coefficient error and admitted noise. It also distinguishes three possible roles of depth—additional address scales, nonlinear forward composition and the cost of transporting a signal along more variable paths.

The experiments walk this hierarchy in order. Controlled simulations separate coordinate bandwidth, neuron-to-tree assignment and task-dependent routing between sibling subtrees; a hierarchy-depth experiment tests the predicted signal–noise crossover; and an experiment that adds physical dendritic stages at fixed synaptic, parameter and state budgets asks when serial nonlinear stages improve forward learning. MICrONS reconstructions then test route capacity, focal shunting and task alignment; an external six-animal dataset tests for signed neuron-specific coordinates. The evidence is conditional: neuron identity supplies most useful information, dendrites offer sparse addresses, shunting regulates gain only in suitable electrotonic regimes, and anatomy helps only when its routes align with task credit (Fig. 1).

Results

We distinguish a global scalar or sign, neuronal identity, within-tree address, and route gain. Theory establishes the transition from point-neuron error to dendritic field; trained networks separate bandwidth, neuron-to-tree ownership and exact transport; anatomical and functional data test candidate addresses, conductance and task alignment. A two-stream credit-reversal task tests routing between sibling subtrees, and a positive-rate hierarchy tests whether serial nonlinear depth contributes at matched resources. Study-design chronology, numerical gates and exclusions are reported together in Methods.

Exact transported compartment error is used only as an information ceiling, never as a proposed biological signal. Every routed advantage reported here is a resource claim rather than a claim of dendrite-exclusive computation: a point model explicitly supplied with the same routing fields and gains matches it by construction, so the question is what anatomy makes available, and at what wiring cost.

Notation used throughout. Indices u, n and i denote a neuron, a compartment and a synapse, respectively. K is the number of communicated feedback channels; H is task-hierarchy depth; D_r is routed address depth in the fixed-state phase experiment; and D_p is physical dendritic stage count in the trained conductance model. The architecture labels D1, D2 and D3 mean $D_\mathrm{p} = 1, 2, 3$. In a label $[b_1, \ldots, b_{D_\mathrm{p}}]$, entries are branching factors ordered from soma to distal tips, and the total number of nonsomatic branch units is $\sum_{j=1}^{D_\mathrm{p}} \prod_{r=1}^{j} b_r$; hence D1 $[8]$ has 8 units, D2 $[2,3]$ has $2+6$, and D3 $[2,1,2]$ has $2+2+4$. The symbols δ_u and $\delta_{u,k}$ denote a neuron-level teaching coordinate and its route-k refinement. We write $\Phi_{u,K}$ for a feedback dictionary and P_Φ for its projector. Bold $\boldsymbol{g} = \nabla\mathscr{L}$ denotes a gradient, whereas scalar g_i denotes a conductance; R_m is reserved for specific membrane resistance. The Supplementary Information contains the complete symbol table.

Exact dendritic gradients factor into eligibility and transported error

For a point neuron u with drive $s_u = \sum_i w_i x_i$, activity $V_u = f(s_u)$ and error $\delta_u = \partial \mathscr{L}/\partial V_u$, the chain rule gives

$$\frac{\partial \mathscr{L}}{\partial w_i} = \underbrace{x_i f'(s_u)}_{\text{synapse-local eligibility}} \underbrace{\delta_u}_{\text{one coordinate shared by neuron } u}. \tag{1}$$

Backpropagation computes δ_u from the downstream Jacobian, whereas biological models approximate it. A point unit has no subcellular state: all its synapses share this coordinate and differ only through eligibility. We ask how a dendritic tree extends that credit.

The local factorization is not restricted to one feedforward tree. Let the steady state of a compartmental neuron satisfy

$$\boldsymbol{F}(\boldsymbol{V}, \boldsymbol{g}, \boldsymbol{x}) = \boldsymbol{0}, \qquad J_V = \frac{\partial \boldsymbol{F}}{\partial \boldsymbol{V}}, \qquad J_V^{\top} \boldsymbol{q} = \nabla_{\boldsymbol{V}} \mathscr{L}. \tag{2}$$

This fixed-point adjoint is the steady-state member of the Almeida–Pineda recurrent-backpropagation lineage [22, 23]. Intuitively, q_n measures how much the loss would change if a small current were injected at compartment n. When J_V is nonsingular, so that the steady state varies differentiably with its parameters, implicit differentiation gives the exact parameter gradient

$$\frac{\partial \mathscr{L}}{\partial g_i} = -\boldsymbol{q}^{\top} \frac{\partial \boldsymbol{F}}{\partial g_i}. \tag{3}$$

For a conductance synapse at compartment n, using the residual convention $\boldsymbol{F} = G\boldsymbol{V} - \boldsymbol{b}$, this becomes

$$\frac{\partial \mathscr{L}}{\partial g_i} = \underbrace{x_i(E_i - V_n)}_{\text{synapse-local factor}} \underbrace{q_n}_{\text{compartment adjoint}}. \tag{4}$$

The directed path product below is the triangular-tree corollary (an acyclic directed tree whose Jacobian can be ordered triangularly) of Eq. 2. In a reciprocal passive cable, $J_V = G$ is symmetric positive definite and $G^{-\top} = G^{-1}$, giving the Green's-function adjoint used for reconstructed cells. For a general nonsymmetric or recurrent compartmental system the inverse transpose is essential. For the triangular tree system used below, $q_n = R_n^{\text{tot}} \partial \mathscr{L}/\partial V_n$, so Eq. 4 and the compartment factorization of Eq. 6 are the same identity with the input-resistance factor moved between the local and transported terms. Thus dendritic credit is always local eligibility multiplied by a compartment-specific adjoint; only the construction of that adjoint changes.

We replace the point unit by a directed, feedforward rooted tree [21] whose compartment n receives synaptic conductances g_i with presynaptic activities x_i and reversal potentials E_i. Let $\mathscr{S}_n$ and $\mathscr{C}_n$ denote its synapses and child compartments, respectively, and let g_n^L and E_L denote leak conductance and reversal potential. At the conductance stage, its steady-state voltage is

$$V_n = R_n^{\text{tot}} \left(\sum_{i \in \mathscr{S}_n} g_i x_i E_i + \sum_{c \in \mathscr{C}_n} g_{c \to n}^{\text{den}} a_c + g_n^L E_L \right), \qquad R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}, \tag{5}$$

where $a_c = f_c(V_c)$ is the activity transmitted by child c. Here $g_n^{\text{tot}} = g_n^L + \sum_{i \in \mathscr{S}_n} g_i x_i + \sum_{c \in \mathscr{C}_n} g_{c \to n}^{\text{den}}$ includes leak, active synaptic conductance, and child coupling. Differentiating with respect to a synaptic conductance gives

$$\frac{\partial \mathscr{L}}{\partial g_i} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \underbrace{\frac{\partial \mathscr{L}}{\partial V_n}}_{\text{compartment error}}. \tag{6}$$

Presynaptic activity, driving force, and input resistance are local to the synapse and its compartment. All remaining task dependence is contained in $\partial \mathscr{L}/\partial V_n$.

Because a tree has one route from each compartment to the soma, its exact compartment error is

$$\frac{\partial \mathscr{L}}{\partial V_n} = \delta_{0,u} \widetilde{\alpha}_n, \qquad \widetilde{\alpha}_n = \prod_{(i \to k) \in \text{path}(n \to 0)} f_i'(V_i) R_k^{\text{tot}} g_{i \to k}^{\text{den}}, \tag{7}$$

where $\delta_{0,u}$ is the somatic error for neuron u. Exact reconstruction from Eqs. 6–7 matched automatic differentiation across 100 diagnostic runs: median relative gradient error was 7.43×10^{-8} and the maximum was 1.88×10^{-7} (Fig. 2a).

Equation 7 applies to the directed artificial network used in the regular-tree experiments. The reconstructed-cell analyses below instead use a reciprocal passive cable matrix grounded in classical dendritic cable theory [24]. In that model the compartment error is an exact Green's-function adjoint rather than this simple product, but the synaptic gradient retains the same separation between local sensitivity and a soma-derived compartment error.

This identity defines what a local learning rule must approximate. Our three-factor rule retains the exact eligibility in Eq. 6 and replaces the compartment error by an available teaching field. Additional four- and five-factor variants use slowly estimated reliability preconditioners, but do not alter the fast local factorization. Exact path transport is used only as an information oracle.

A stochastic credit operator defines when dendritic structure can help

The factorization identifies the object that differs among local rules. Let $\widehat{\boldsymbol{g}} = \boldsymbol{g} + \boldsymbol{\xi}$ be an unbiased stochastic gradient with covariance Σ, and let a fixed credit operator M encode the available coordinate, route assignment and gain. The update is $\boldsymbol{w}^{+} = \boldsymbol{w} - \eta M \widehat{\boldsymbol{g}}$. If the population loss has L-Lipschitz gradient, the descent lemma gives

$$\mathbb{E}\mathscr{L}(\boldsymbol{w}^{+}) \leq \mathscr{L}(\boldsymbol{w}) - \eta \boldsymbol{g}^{\mathsf{T}} M \boldsymbol{g} + \frac{L\eta^2}{2}\left(\|M\boldsymbol{g}\|_2^2 + \operatorname{tr}(M\Sigma M^{\mathsf{T}})\right). \tag{8}$$

When $\boldsymbol{g}^{\mathsf{T}} M \boldsymbol{g} > 0$, optimizing this bound over η yields the phase utility

$$U(M) = \frac{[\boldsymbol{g}^{\mathsf{T}} M \boldsymbol{g}]^2}{2L\{\|M\boldsymbol{g}\|_2^2 + \operatorname{tr}(M\Sigma M^{\mathsf{T}})\}}. \tag{9}$$

Equation 8 is an upper guarantee for a general L-smooth loss and is exact for the corresponding quadratic when its curvature equals L on the update directions; comparisons of $U(M)$ otherwise order guarantees, not necessarily realized losses. Thus more address dimensions are predicted to help only when their retained task signal outweighs finite-step gain and admitted gradient noise. Exact full-batch backpropagation has $M = I$ and $\Sigma = 0$; at a matched infinitesimal update norm no alternative direction can provide greater first-order descent. A projected stochastic gradient can nevertheless outperform an unprojected stochastic gradient because it rejects noise. For an orthogonal projector P on an identity-curvature quadratic at the common step $\eta = 1/L$, the exact crossover is $\operatorname{tr}[(I-P)\Sigma] > \|(I-P)\boldsymbol{g}\|_2^2$: rejected noise must exceed discarded signal (at smaller common steps the rejected-noise threshold rises; Supplementary Information). Backpropagation followed by the same projection is therefore the relevant upper control for any projected local rule.

Scope of the credit-operator prediction. One-step geometry predicts within-family progress and large contrasts caused by retained signal, rejected noise or gross misalignment. It does not predict every advantage accumulated along a training trajectory; in particular, the trained $K = 4$ ancestry advantage is outside the initialization bound's scope.

Capture vocabulary. *Field capture* is the fraction of exact mean-field energy retained by a dictionary projection; *spectral capture* averages this fraction over a credit covariance; *checkpoint capture* applies it to a trained gradient; and *capture per wiring* divides field capture by feedback-connection density. We use an explicit qualifier whenever the estimand is not field capture.

For a route dictionary Φ, topology controls the span $\operatorname{col}(\Phi)$, whereas learning must also estimate the route coefficients. The total weighted field error separates exactly into the irreducible projection residual and the within-span coefficient-estimation error (Supplementary Information, "Stochastic credit-operator phase theory"). Multiplying fixed route columns by nonzero scalar shunting gains leaves their span and spectral capture unchanged, although it can strongly alter Gram conditioning. By contrast, state- or branch-dependent gain can act as a reliability preconditioner. For disjoint branches with signal energy S_b, noise energy N_b and fixed step $\eta = c/L$, Eq. 8 is

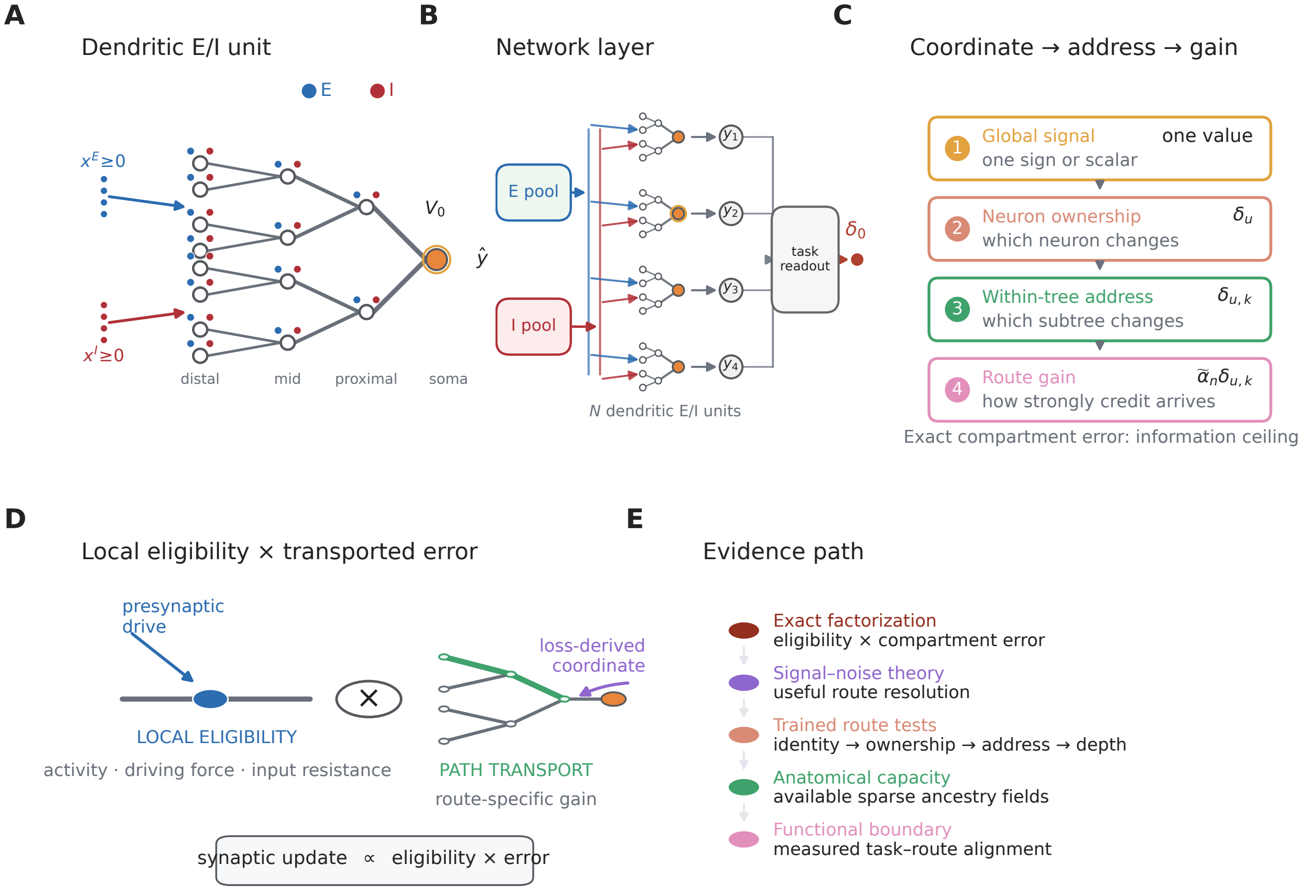

Figure 1: **From neuronal coordinates to dendritic credit addresses.** **A**, Local synaptic states are distributed across a branching tree, whereas the task supplies an error at the neuronal output. **B**, A population layer of dendritic excitatory and inhibitory units supplies a task readout; output error first arrives at each neuronal soma. **C**, Four distinct levels of spatial information: a global signal, neuron-to-coordinate ownership, within-tree address, and route gain. The following experiments test these levels in that order before asking when additional route resolution is useful. **D**, The central factorization as a teaching cartoon: local synaptic eligibility is multiplied by a neuron-level error transported through a route-specific gain. **E**, Evidence path from factorization and signal–noise theory through trained route tests, anatomical capacity and the functional alignment boundary.

optimized over attenuations $0 \leq a_b \leq 1$ by $a_b^* = \min\{1, S_b/[c(S_b+N_b)]\}$; the reliability factor $S_b/(S_b+N_b)$ is the special case $c = 1$, paralleling the forward reliability weighting derived for conductance-based dendrites [25]. A point model supplied with the identical branch gates must match; the biological question is whether dendritic conductance can make such localized gain available.

Per-neuron bandwidth dominates restricted feedback

A global scalar is low bandwidth but discards which neuron produced an error. The matched-width/scalar-fallback rule used in our initial experiments retains the somatic vector only at a dendritic stage of identical width and otherwise averages it to one scalar per example. In contrast, neuron-indexed feedback repeats the appropriate somatic coordinate across all compartments belonging to that neuron's tree. It communicates one value per neuron, not one value per compartment.

This scalar-versus-indexed distinction is a feedback-bandwidth effect, not a new scaling principle: global perturbation estimates are known to become noisy as network size grows [10, 26]. Our question is what is added by neuron-to-tree ownership and within-tree address after neuronal identity has been supplied.

We compared these fields in regular $[3,3]$ trees, which contain 13 modeled compartments per neuron. Across 15 paired MNIST seeds, neuron-indexed feedback raised three-factor accuracy from $90.54 \pm 0.54\%$ to $97.16 \pm 0.11\%$ in shunting networks and from $91.76 \pm 0.78\%$ to $97.14 \pm 0.11\%$ in width- and connectivity-matched additive networks. All 15 pairs improved in each architecture. At the same neuron-indexed-trained checkpoints, the cosine between local and exact dendritic gradients rose from 0.012 ± 0.059 to 0.708 ± 0.027 in shunting trees and from -0.001 ± 0.047 to 0.625 ± 0.035 in additive trees (Fig. 2b,c). The increase was positive at all 15 matched checkpoints in each architecture.

Supplying exact transported compartment errors tests the derived local eligibility without feedback approximation. With five paired shunting seeds, three-factor learning with a local decoder reached $97.18 \pm 0.12\%$, compared with $97.03 \pm 0.08\%$ for a same-architecture backpropagation reference. Adding the empirical five-factor preconditioner or a backpropagated decoder provided no material gain under exact transport (Fig. 2d). Thus the poor scalar-feedback result is not a failure of the derived local eligibility. The large scalar-to-neuron-indexed gap results from loss of neuronal identity; the smaller remaining gap to exact transport reflects omitted path information.

Shunting changes the conductance component of path gain. In a separate five-seed diagnostic cohort with five inhibitory synapses per branch, the unweighted path-gain coefficient of variation across compartments within examples was 0.229 for shunting and 1.161 for additive trees, with the same ordering in all five paired seeds. This statistic describes the electrical operating point; it does not establish better gradient direction. Indeed, a 15-seed comparison crossing architecture with analytical or occupancy-calibrated branch-output-nonlinearity initialization showed no general shunting advantage: both architectures reached a mean of 90.92% under analytical initialization (± 0.42 and ± 0.51 s.d. for shunting and additive; paired difference -0.00, $P = 0.982$), whereas occupancy calibration modestly favored the additive model. We therefore interpret conductance as a route-gain mechanism, not as a universal optimizer (Fig. 2e,f).

Broader archived controls define the boundary of this conclusion (Supplementary Figs. S1–S4). Under restricted feedback, shunting trees were more stable than matched additive trees in depth and teaching-noise stress tests, but the advantage was task dependent and did not isolate a backward shunting mechanism. Increasing feedback structure consistently improved learning, including on a flattened CIFAR-10 mechanism control, whereas unmatched local errors and forward nonlinearities did not replace the required teaching field. These experiments support a conditional robustness claim rather than a general accuracy advantage.

We prospectively trained 640 models crossing two tasks, two neuron models, one to four dendritic stages, three feedback fields and matched backpropagation, with ten paired seeds per condition; all local conditions used the same factorization-derived eligibility and training protocol. An input-validity audit, defined without reference to outcomes, excluded all 160 synthetic-noise shunting runs, whose signed transfer output is not a valid drive for a positive-conductance shunting denominator (Methods), leaving 480 retained runs comprising both models on MNIST and the additive model on the noise task.

Neuron-indexed feedback increased held-out accuracy over a scalar broadcast by 4.92–6.99 percentage points on MNIST and 9.76–14.16 in the additive noise model, with all 120 retained paired seeds improving (Fig. 2g,h). Exact transport placed mean three-factor accuracy within 0.14 points of matched backpropagation in all 12 retained task–model–depth conditions (Fig. 2i).

A fresh Fashion-MNIST ladder tested whether the dominant identity effect was specific to MNIST (Fig. 2p,q). Without retuning the architecture or learning protocol, neuron-indexed feedback improved over the scalar fallback by 4.46 points in shunting trees (95% paired-seed interval 4.09–4.88) and 3.78 points in additive trees (3.43–4.14), with all ten paired seeds positive in both architectures (exact sign-flip $P = 0.002$). Exact path transport added no reliable benefit beyond neuron identity: -0.11 points in shunting trees (-0.21 to -0.01; 2/10 positive; $P = 0.096$) and 0.13 points in additive trees (-0.08 to 0.35; 8/10 positive; $P = 0.314$). This second dataset strengthens the bandwidth conclusion while leaving the distinct within-tree-address question to the credit-reversal experiments below.

A bandwidth-matched control fixed coordinate count and values but deranged the coordinate-to-tree map. Across the six retained conditions, correct assignment improved accuracy by 0.15–0.70 points and was favored in 58/60 pairs (Fig. 2j); five of six contrasts survived correction. Bandwidth therefore explains most of the scalar-to-neuron-indexed gain, while correct coordinate ownership adds a smaller effect. This comparison does not assign different task-dependent coordinates to different subtrees of one neuron and is not a trained test of within-tree addressing.

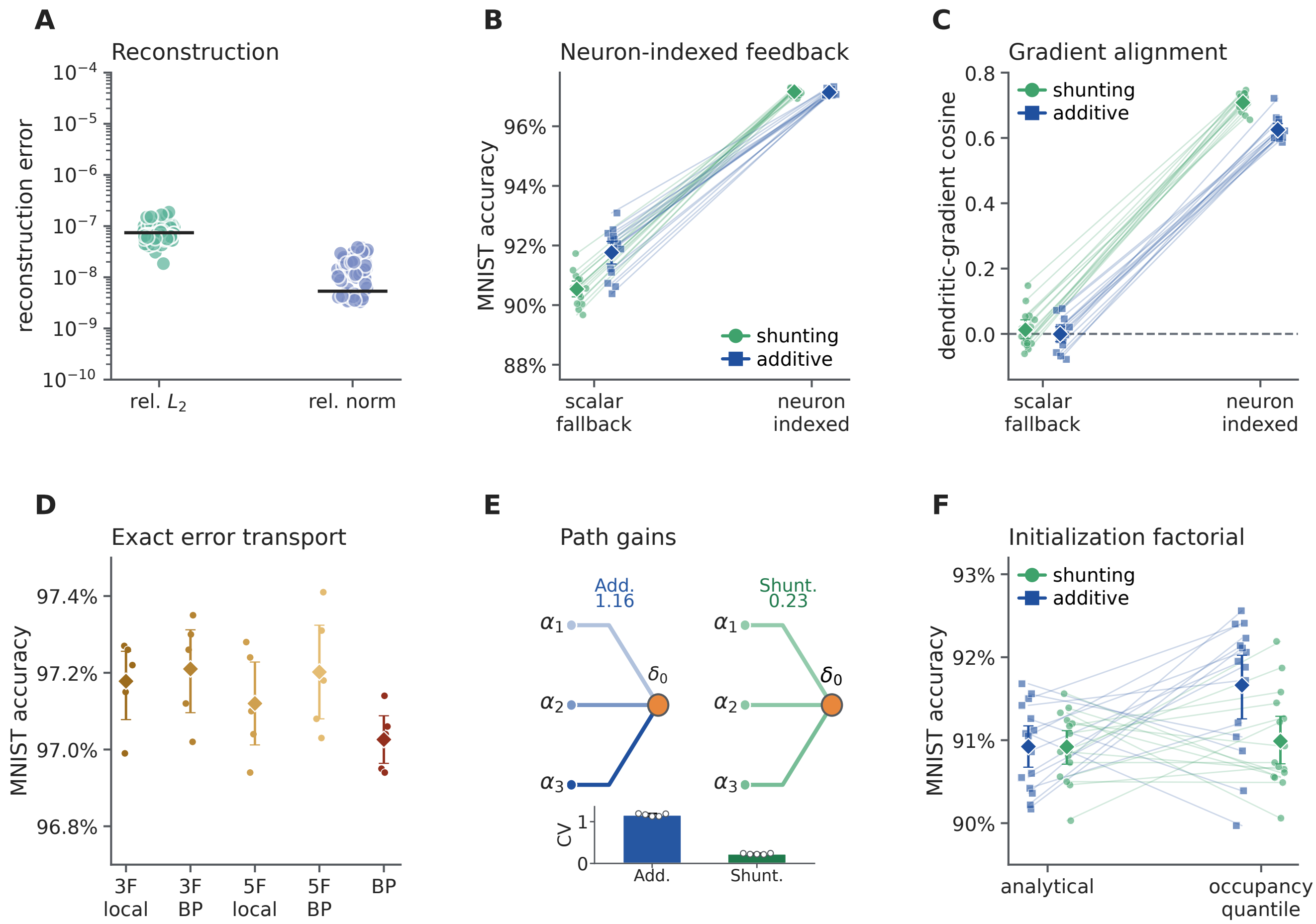

Figure 2: **Feedback identity rather than the local eligibility limits learning in regular trees. A**, Relative error between the analytic factorization and automatic differentiation in 100 diagnostic runs, shown as relative L_2 error and relative gradient-norm error; lines mark medians. **B**, Three-factor MNIST accuracy under scalar-fallback and neuron-indexed feedback for 15 paired seeds in each architecture. **C**, Dendritic-gradient cosine when the two feedback fields are evaluated at the same neuron-indexed-trained checkpoints ($n = 15$ seeds per architecture). **D**, Accuracy under exact path transport for three- and five-factor rules with local or backpropagated (BP) decoder updates ($n = 5$ seeds); the rightmost BP condition is the same-architecture backpropagation reference. **E**, Path-gain coefficient of variation in five paired shunting and additive checkpoints. This is an unweighted operating-point statistic, not a gradient-direction metric. **F**, Accuracy in the 2×2 architecture-by-branch-output- initialization factorial ($n = 15$ seeds per condition). Lines in **B,C,F** connect identical seeds. Large points and intervals in **B–D,F** show means and 95% bootstrap intervals; smaller points are individual seeds. Panels **D,F** reuse cohorts of the original regular-tree study [21]; panels **B,C** report a re-execution of the feedback-definition experiment under the released implementation with the width- and connectivity-matched additive architecture (Methods). Exact transport is an oracle, not a proposed biological signal.

We therefore froze a separate two-stream credit-reversal task before its outcomes were inspected. Both sibling streams were active on every example; context selected the stream controlling the somatic output, while the other carried an anticorrelated distractor. Context and label were exactly balanced, so context alone could not predict the target. The same scalar somatic error therefore required different branch updates in paired contexts. The one-neuron field captured 0.509 of the eligibility-weighted exact-gradient energy at initialization, below the prespecified 0.90 rejection threshold, whereas two correct subtree coordinates captured 1.000.

Across ten paired seeds, correct two-subtree routing reached 80.63% held-out accuracy, compared with 76.24% for one shared neuronal coordinate (difference 4.39 percentage points, 95% paired bootstrap interval 3.33–5.48;

10/10 seeds) and 19.63% after a fixed within-neuron derangement (a permutation assigning every route to the wrong sibling) (difference 61.00 points, 59.93–62.09; 10/10 seeds; Fig. 2m). A dense random rank-two map reached 58.47% and was below correct routing in all ten seeds. Gradient capture tracked norm-matched one-step progress (Fig. 2n). After a single-context adaptation block, the shared rule lost 55.66 points on the previous context, whereas the correct route left the inactive sibling unchanged (difference in forgetting, 55.66 points, 53.84–57.67; Fig. 2o).

Correct subtree routing, exact transport and analytic backpropagation were identical on this shallow linear branch model. Crucially, a point-neuron implementation with the same two context-gated parameter groups also matched them exactly. The experiment therefore demonstrates the value of the address, not a unique dendritic computational advantage: the same solution can be paid for by explicit feature gating in a point model. It executes the decisive $K = 2$ support contrast on one forward architecture. We next completed the broader bandwidth and representation factorial.

The same audit excluded the 400-run inhibitory-dose family from inference because its shunting-minus-additive endpoint depends on the invalid signed-shunting cells. An independent clean-source 320-run exact-transport/backpropagation audit instead tested implementation agreement across ten paired seeds, four depths, two tasks and two cores; the synthetic shunting cells used an explicit ReLU transfer before positive conductances were formed (Fig. 2k). All 320 runs passed the configuration, log, metric and finite stage-complete checkpoint audit. After averaging depths within seed, exact-minus- backpropagation accuracy was 0.005 percentage points on additive MNIST (95% paired bootstrap interval -0.028–0.041), -0.013 on shunting MNIST (-0.047–0.022), -0.112 on additive noise (-0.465–0.247) and -0.098 on shunting noise (-0.653–0.390). All four intervals included zero. Across all 160 raw method pairs, the mean difference was -0.055 points and the largest absolute pair difference was 5.90 points. These are empirical agreement checks, not formal equivalence tests or pointwise identity claims.

A retained 240-run spatial-map control improved accuracy by 0.36–2.57 points, including under backpropagation and in the additive randomly projected noise model (Supplementary Fig. S7). At 336 contacts, spatial branches covered 336 unique inputs versus 276.2 for random branches, identifying a forward-connectivity rather than task-specific credit-routing effect.

Of 320 fixed-budget runs, the 160 additive cells provide the valid synthetic comparison at 16 terminal branches and 960–968 contacts per soma. Depth four underperformed depth one in all 40 paired seed contrasts: -7.92 points under scalar feedback, -1.06 under neuron-indexed feedback, and -1.37 under both exact transport and backpropagation (Fig. 2l; Supplementary Fig. S8). Deeper trees retain more trainable parameters, but scalar feedback clearly amplified depth stress.

At 120 retained backpropagation checkpoints, mean gradient cosine was 0.027 for scalar and 0.586 for neuron-indexed feedback. Norm-matched updates descended at 75/120 and 120/120 checkpoints, respectively, and cosine predicted retained one-step progress ($\rho = 0.893$, 95% checkpoint-bootstrap interval 0.860–0.923; Supplementary Fig. S9). This connects field geometry to loss change without replacing the trained-learning factorial.

Structured subtree addresses help at an intermediate feedback budget

We froze an eight-context hierarchical credit-reversal task and crossed five parameter-matched representations, feedback budgets $K \in \{1,2,4,8\}$, six route families and three references. Twenty paired independent seeds produced all 2,700 planned fits. Context and label were exactly balanced; every route field had unit norm per example; initial weights, data and optimization were paired. The dendritic, explicitly gated point, flat-compartment and grouped-point implementations had 64 forward parameters, 64 active input contacts and one nonlinearity. They received identical routed coefficient fields. A degree- and depth-matched rewired tree preserved resource counts while breaking task–topology alignment.

Correct ancestry routing displayed a non-monotonic advantage. Mean held-out accuracy increased from 0.187 at $K = 1$ to 0.449, 0.801 and 0.810 at $K = 2,4,8$, respectively (Fig. 3b). It strongly exceeded fixed within-neuron route derangement at $K = 2$ (difference 0.264, 95% paired bootstrap interval 0.243–0.285) and $K = 4$ (0.611, 0.601–0.620). Against the best matched non-anatomical control selected separately within each seed, however, correct ancestry lost at $K = 1$ and $K = 2$, won at $K = 4$ by 0.0127 (0.0059–0.0197; 15/20 seeds; two-sided Wilcoxon $P = 0.0048$, Benjamini–Hochberg-adjusted $P = 0.0064$ across the four budgets), and tied at full rank $K = 8$ (Fig. 3c). An

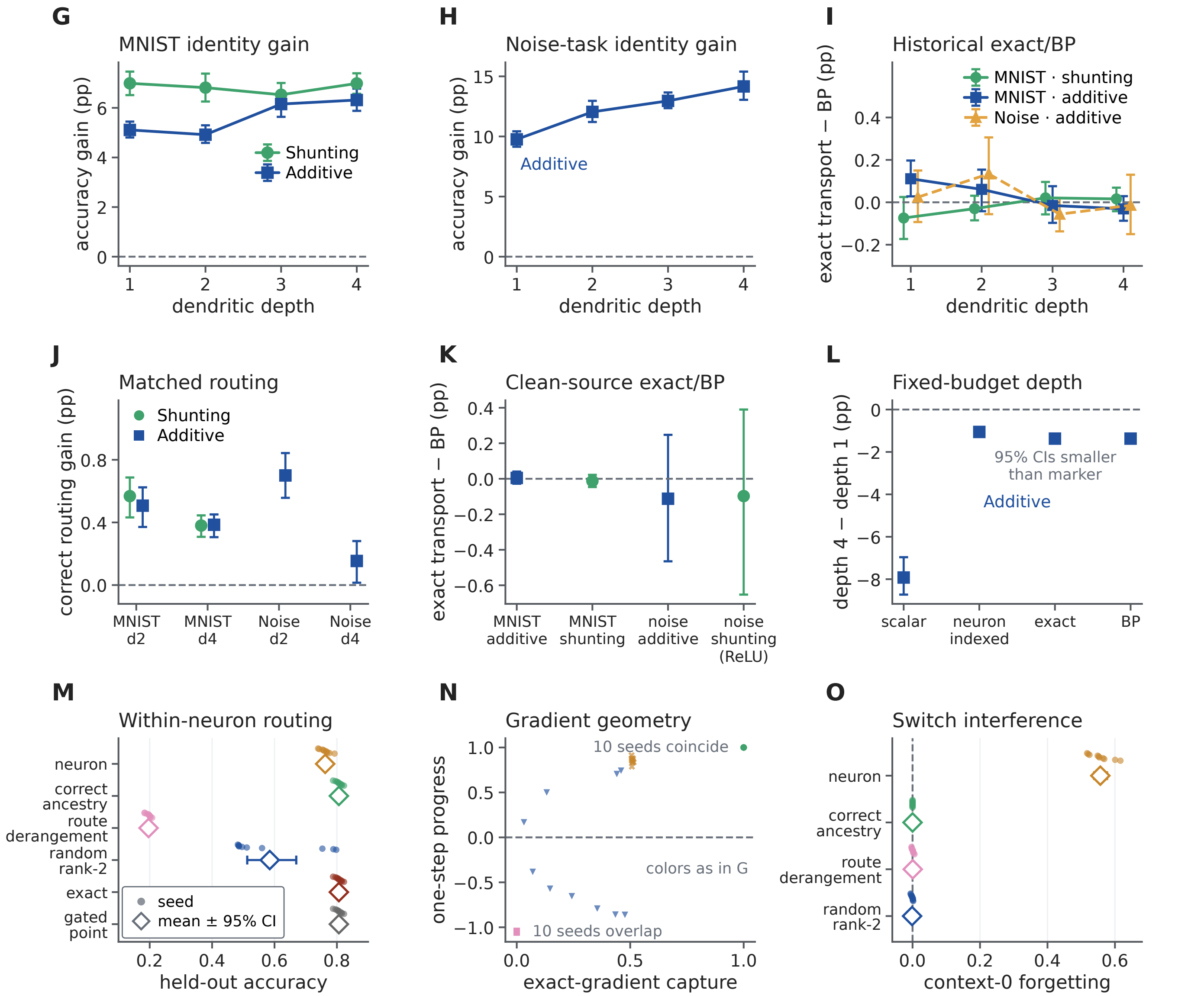

Figure 2: **Prospective trained tests resolve coordinate, ownership and subtree-address effects. G,H**, Neuron-indexed-minus-scalar accuracy across depth on MNIST and in the input-valid additive noise model. **I**, Exact-transport-minus-backpropagation accuracy in the retained historical conditions (solid circles and squares, MNIST; dashed triangles, noise task; green and blue identify the forward core as in **G**). **J**, Gain from correct rather than fixed- deranged neuron-to-tree ownership at matched bandwidth. **K**, Seed-level depth-averaged exact-minus-backpropagation accuracy in the independent clean- source audit; synthetic shunting uses a non-negative transfer and every interval includes zero. **L**, Additive-model depth-four-minus-depth-one accuracy after matching terminal branches and input contacts. **M**, Held-out two-stream credit-reversal accuracy; correct subtree, exact transport and gated-point values coincide. **N**, Exact-gradient capture versus norm-matched one-step loss reduction; colors identify the first four fields in **M**. **O**, Context-0 forgetting after context-1-only adaptation. Small points are paired seeds; larger symbols and bars are means and 95% seed-bootstrap intervals ($n = 10$). Panels **M–O** test within-neuron support at $K = 2$ on one matched forward architecture. Exact transport is an information upper bound.

unrestricted learned rank-K oracle and random rank-K routes therefore exploited low-dimensional task structure more efficiently at the smallest budgets; nested anatomy became advantageous only when four coordinates were available.

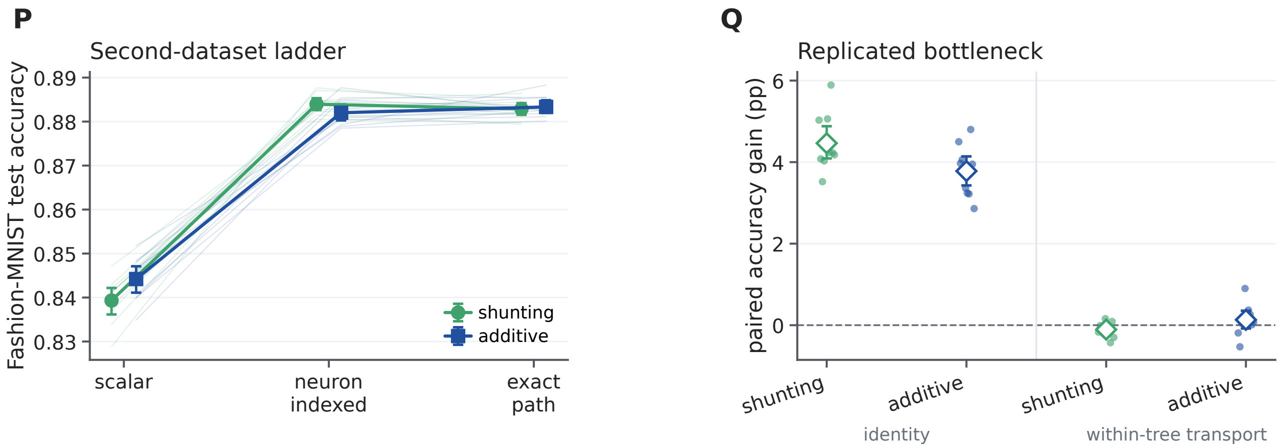

Figure 2: **The neuronal-identity bottleneck replicates on Fashion-MNIST. P**, Held-out accuracy under scalar fallback, neuron-indexed feedback and exact path transport in shunting and additive trees. Faint lines connect identical seeds. **Q**, Paired accuracy gains for neuron identity over scalar fallback and exact path over neuron identity. Large diamonds and bars show means and 95% paired-seed bootstrap intervals; small points are the ten paired seeds. The architecture, training protocol and hyperparameters were inherited without retuning from the MNIST ladder. Exact path transport is an oracle.

The correct matched tree exceeded the rewired tree by 0.231 (0.210–0.254) at $K = 2$ and 0.050 (0.042–0.060) at $K = 4$, with all 20 seeds positive in both contrasts (Fig. 3d). At $K = 1$ the route is shared, and at $K = 8$ every leaf is addressed, so rewiring had no effect. Exact transport and analytic backpropagation were identical. The four implementation-equivalent dendritic, gated-point, flat and grouped-point controls also matched exactly in every deterministic and paired-stochastic condition (maximum accuracy difference 0; Fig. 3e). Initial exact-gradient capture alone did not determine final accuracy across route families: fields with similar capture but different route signs and task alignment learned differently (Fig. 3f). Thus the result establishes a structured-address resource advantage at an intermediate bandwidth, not an intrinsic superiority of dendritic material: point models reproduce it when explicitly supplied with the same parameter groups and routing fields.

A credit-operator phase theory predicts when restricted routes help

We next froze a controlled experiment directly from Eqs. 8–9. Sixteen coordinates were represented in an orthonormal balanced-tree Haar basis. Fifty independent confirmatory seeds generated 10,800 seed–condition rows crossing route rank, task–tree alignment, task hierarchy, routed model depth, signal and noise retention, and branch-reliability heterogeneity. The experiment holds the forward coordinate budget fixed and isolates stochastic credit geometry; it is not a positive-conductance forward-neuron simulation.

First, the benefit of ancestry routes depended continuously on spectral task–tree alignment $\rho \in [0, 1]$, where $\rho = 0$ is the seed-specific unaligned covariance and $\rho = 1$ is tree-Haar-aligned. At full alignment and $K = 4$, ancestry captured 0.552 more task-gradient energy than a random rank-matched route (95% paired bootstrap interval 0.535–0.569; 50/50 seeds; two-sided Wilcoxon $P = 1.78 \times 10^{-15}$; Fig. 4b). The advantage was near zero under random alignment, increased with alignment, never exceeded the Ky–Fan dense principal-subspace upper bound—the maximum covariance capture available to any rank-K orthogonal projector—and vanished at full rank. At full alignment and $K = 4$, ancestry attained the rank-matched upper bound to numerical precision. This attainment is the generic case for hierarchical credit: the K-route ancestry span equals the span of the K coarsest tree-Haar modes, so anatomy attains the rank-K Ky–Fan bound whenever the task covariance is diagonalized by the tree's wavelet basis with a coarse-dominant spectrum—and, in general, exactly when the leading rank-K eigenspace lies inside the laminar span—while its shortfall for a general covariance is bounded by the leading-eigenspace energy outside that span (Supplementary Section S4). A deterministic endpoint reanalysis used the affine covariance mixture to obtain an exact ancestry-versus-random alignment threshold within each seed. Nineteen of 50 random-route draws

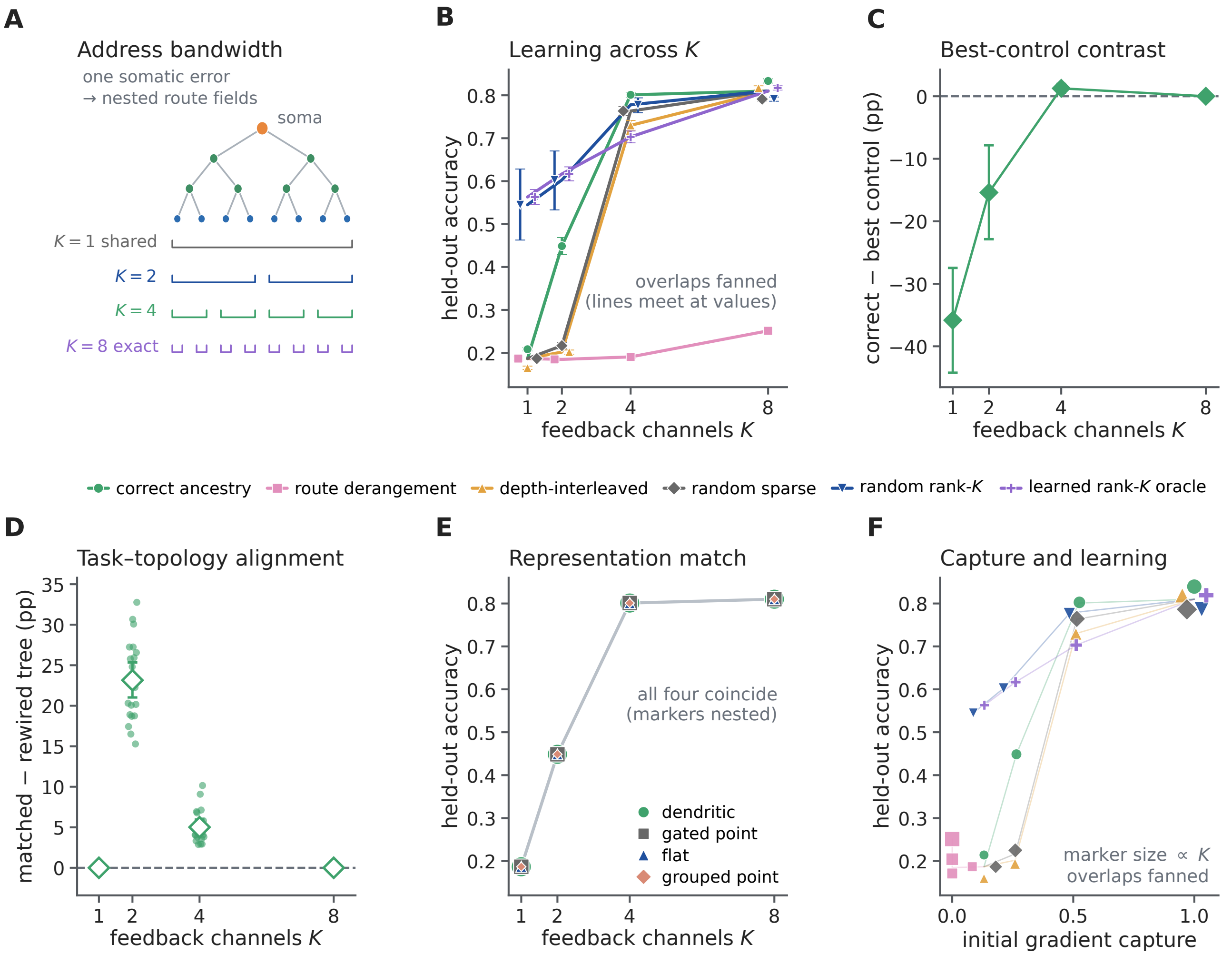

Figure 3: **The benefit of a structured subtree address depends on feedback bandwidth and task–topology alignment.** **A**, Eight leaves are addressed by one shared coordinate or $K = 2,4,8$ nested route fields. **B**, Held-out accuracy across route families in the dendritic-tree implementation. Curves and error bars show means and 95% seed-bootstrap intervals over 20 paired seeds; overlapping points are fanned slightly, with lines meeting at the shared values. The learned rank-K condition is an unrestricted task-derived upper bound. **C**, Correct ancestry minus the best of four matched non-anatomical controls within each seed; positive values favor ancestry. **D**, Correct routing on the task-matched tree minus the same rule on a degree- and depth-matched rewired tree. Small points are paired seed differences. **E**, Correct-route learning is numerically identical for dendritic, explicitly gated point, flat-compartment and grouped-point implementations when each receives the same routed fields. **F**, Initial exact-gradient capture versus final accuracy for each route family and budget; marker size increases with K, and overlapping points are fanned for visibility. Capture alone does not determine learning when route signs and task structure differ. Every condition matched forward parameter count, active contacts, nonlinearity count, initialization, data and optimizer.

already favored ancestry at zero alignment; all 50 favored it at full alignment. Counting the former as requiring zero alignment, the median $K = 4$ threshold was 0.0173 and the mean was 0.0523 (95% seed-bootstrap interval 0.0336–0.0730). This is a threshold relative to sampled random routes, not a universal biological constant.

Second, routed depth was optimal when it matched task hierarchy. This match was imposed by the controlled generator—signal occupied levels through H, whereas finer levels carried additional noise—so the test is a quantitative validation of predicted crossover locations, not a discovery that $D_r = H$ in arbitrary tasks. For task depths $H = 1,2,3,4$, the lowest mean final population loss occurred at model depth $D_r = H$ (Fig. 4c,d). Across the

200 seed–task pairs, matched depth reduced loss relative to the best mismatched depth by 0.0957 (95% interval 0.0782–0.1134 after averaging the four task-depth contrasts within each independent seed; 48/50 seeds; two-sided Wilcoxon $P = 3.00 \times 10^{-13}$). Insufficient depth left target components outside the route span; excess depth admitted fine-scale stochastic gradient energy. At $D_\mathrm{r} = 4$ the projector is full rank and the update is identical to the common stochastic-backpropagation reference. Depth therefore helps here by matching the address resolution of the task, not by being large in isolation.

Third, independently varying signal and noise retention recovered the predicted boundary. Backpropagation followed by the same route projection had lower mean one-step population loss than unprojected stochastic backpropagation in exactly those of the 25 condition means for which rejected noise exceeded discarded signal (Fig. 4e). The routed local estimator carried an additional specified coefficient-noise cost and never exceeded this projected-backpropagation control. Branch-specific reliability gains then improved progress only when they were aligned with branch signal-to-noise ratio. At maximum heterogeneity, reliability-aligned gain exceeded the best single global gain by 1.241 in one-step loss reduction (95% interval 1.116–1.373; 50/50 seeds; $P = 1.78 \times 10^{-15}$), whereas shuffling or reversing the gains removed or reversed the benefit (Fig. 4f). An explicit point gate receiving the same gains matched exactly (maximum difference 0).

We next tested whether a physical conductance could realize this oracle attenuation in a trained branch model. Eight branches received strictly nonnegative input rates and excitatory conductances. A nonnegative shunt to reversal zero reduced branch input resistance, while an oracle compensating current $\kappa_b V_b$ held the forward voltage identical; this isolates the gain multiplying local eligibility. Fifty fresh confirmatory seeds crossed five reliability-heterogeneity levels and nine matched controls (Supplementary Fig. S13). At maximum heterogeneity, reliability-aligned shunting increased one-step test-loss reduction relative to the best global shunt by 0.001817 (95% paired bootstrap interval 0.001618–0.002020; 49/50 seeds; two-sided Wilcoxon $P = 3.55 \times 10^{-15}$). Its mean one-step advantage over no shunt was 0.001339 (0.000594–0.002096; 33/50 pairs; $P = 0.0030$).

After 40 updates, aligned shunting reduced test loss by 0.02040 relative to the best global shunt (-0.02222 to -0.01871 for aligned minus global; 50/50 seeds favored aligned, $P = 1.78 \times 10^{-15}$), and also outperformed shuffled and anti-aligned shunts. Its final loss differed from the unshunted noisy rule by 0.000425 (-0.002456–0.003071; $P = 0.154$), providing no reliable final-loss advantage in either direction. Exact clean backpropagation remained the lowest-loss reference. A state-matched additive control was identical to no shunt, and the explicit point gate receiving the same sample-dependent factors was computed through an independent update path and agreed with the conductance trajectory to numerical precision. Thus positive conductance approximates the predicted reliability ordering under a state clamp, but the immediate benefit does not establish autonomous shunt learning, dendrite-exclusive computation or uniform training superiority.

Finally, static nonzero gain rescaling changed numerical conditioning without changing the route span: nested indicators, their tree-Haar basis and a gain-scaled version had identical capture to 1.33×10^{-15}, while the condition number ranged from 1 for the Haar basis to 583 for the illustrated scaled parameterization (Fig. 4g,h). We then prospectively trained noisy route coefficients in a separate 50-seed, exactly same-span experiment (Supplementary Fig. S14). The three rank-eight coordinates had condition numbers 1, 15 and 388.5 and zero target address residual. The preregistered unconditional Haar advantage was falsified at effective sample size four: raw nested and scaled nested coordinates reduced final loss by 0.277 and 0.191 relative to Haar because their slow modes filtered update noise. At effective sample size 256 the ordering reversed, with Haar reducing loss by 0.0256 and 0.1709, respectively, in 50/50 paired seeds. An exact finite-time bias–variance calculation predicted median crossovers at effective sample sizes 38.5 and 8.25, and an oracle Gram-preconditioned control made every same-span field trajectory identical to 1.16×10^{-15}. Thus conditioning is neither a pure defect nor new capacity: it trades finite-time bias against estimator variance.

The phase utility was also diagnostic in the already completed 2,700-fit factorial. Reconstructing 64 initialization minibatches for each of 20 seeds gave 540 dendritic-route conditions; $U(M)$ predicted norm-matched one-step progress with Spearman $\rho = 0.937$ (seed-block 95% interval 0.9336–0.9412) and final held-out accuracy with $\rho = 0.916$ (0.905–0.929). Spectral capture alone predicted one-step progress less well ($\rho = 0.877$), with a positive seed-block difference of 0.060 (0.050–0.070; Fig. 4i). This secondary reanalysis explains immediate geometry at the fixed initialization; it is not an independent trained experiment.

The bound also locates optima, with an instructive boundary (Supplementary Fig. S16). Because every tested field is a projector on one nested ladder, utility rises with feedback bandwidth only while each added route scale contributes proportionally more signal than admitted noise, so an interior optimum is predicted whenever the signal spectrum decays faster than the noise (Supplementary Section S4). In the depth experiment, the per-seed argmax of the initialization utility coincided with the trained loss minimum in 135 of 200 seed–task pairs, and the group-mean argmax matched exactly at $H = 1,2,3$; at the full-rank boundary $H = 4$ the one-step bound was conservative, favouring $D_{\mathrm{r}} = 2$. In the factorial, the same utility reproduced the exact ties at $K = 8$, where all spans coincide, but did not predict the trained $K = 4$ ancestry advantage over the best control (6 of 20 seeds): that advantage accrues along the training trajectory and is not visible in one-step initialization geometry, consistent with capture alone not determining learning (Fig. 3f).

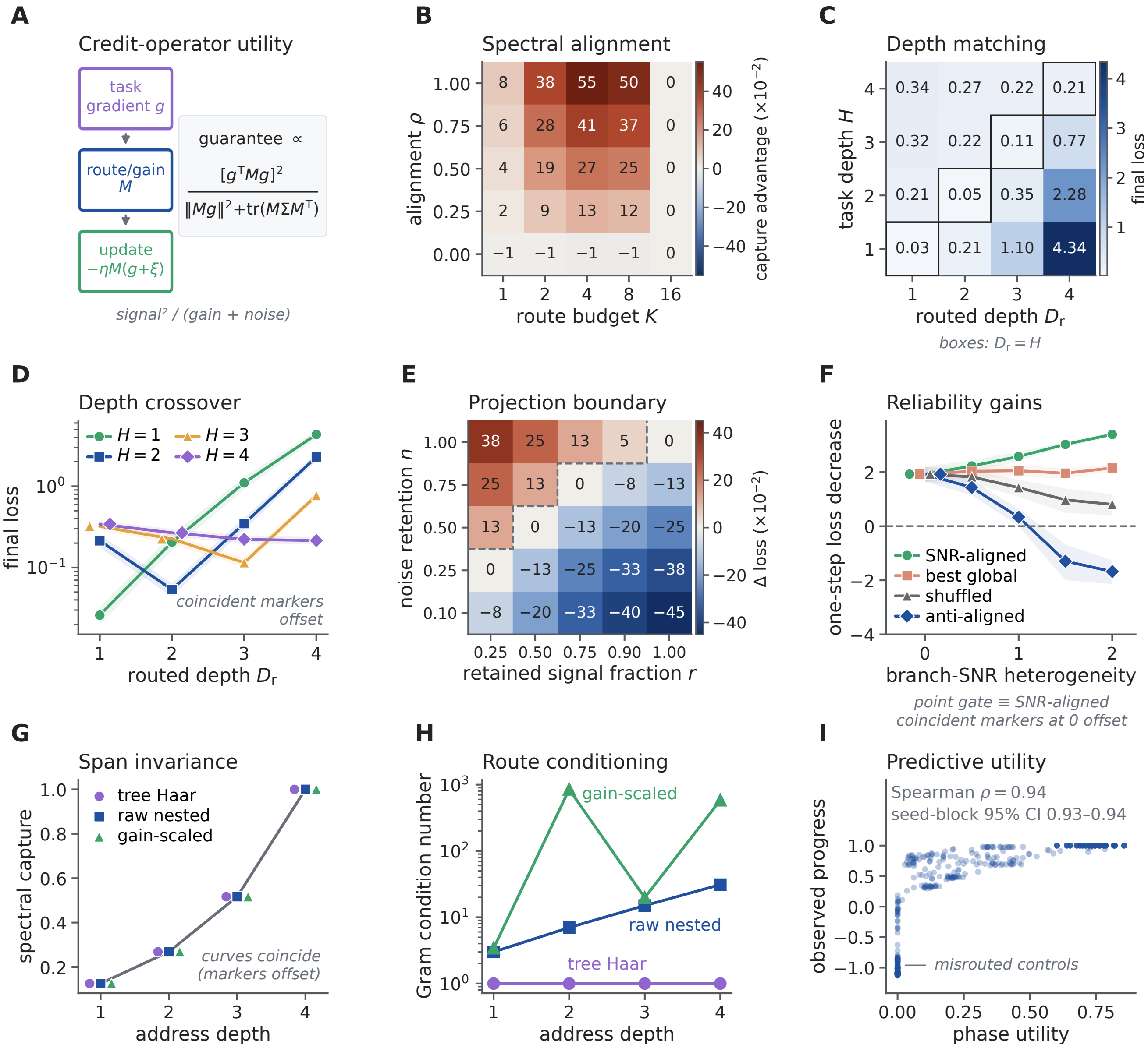

Figure 4: **A signal–noise phase theory predicts when dendritic credit structure is useful.** **A**, A fixed credit operator M maps an unbiased stochastic gradient into a routed local update; its guaranteed utility balances retained signal against finite-step gain and admitted noise. **B**, Mean ancestry-minus-random spectral capture across alignment and rank ($n = 50$ seeds); the advantage grows with alignment and vanishes at full rank. **C,D**, Mean final population loss for crossed task hierarchy H and routed depth D_r. Black outlines mark $D_r = H$; shading in **D** is the 95% seed-bootstrap interval, and coincident markers in **D** are slightly offset. $D_r = 4$ is full rank and equals stochastic backpropagation. **E**, Expected one-step loss of backpropagation plus the route projection minus unprojected stochastic backpropagation. Negative values favor projection; the dashed line is the analytical $n = r$ crossover. **F**, Simulated one-step loss decrease for reliability-aligned, best global, shuffled and anti-aligned gains (means and 95% intervals over 50 seeds). The explicit point gate is identical to the aligned condition, with coincident markers offset at zero heterogeneity. **G,H**, Static nonzero column gains preserve spectral capture but can alter the positive-spectrum Gram condition number. Markers in **G** are horizontally offset to reveal exact overlap. **I**, Prespecified signal–noise–curvature utility versus observed norm-matched one-step progress for 540 dendritic conditions from the 2,700-fit factorial; the interval resamples 20 seed blocks. Panels **B–H** are a fixed-state stochastic quadratic validation, not a positive-conductance forward model or evidence of uniform superiority to exact full-batch backpropagation.

Task-aligned physical depth improves learning at fixed budgets

The fixed-state phase experiment tests address resolution, not whether serial dendritic dynamics contribute to forward computation. We therefore constructed a separate positive-rate task—all inputs are nonnegative rates driving positive conductances—in DendriNet, the conductance-based population-network implementation used throughout (Methods). A distal class signal was modulated by nested fine, coarse and global gains, while separate nonnegative inhibitory streams measured those factors. This is a mechanism-matched existence test built around divisive normalization, not a generic task benchmark [27, 28]. D1 [8], D2 $[2,3]$ and D3 $[2,1,2]$ contained the same eight nonsomatic branch units per soma, 14,336 active contacts, 21,760 candidate slots, 66,178 trainable parameters and 2,944 persistent state scalars. Only the number and placement of serial divisive stages changed (Fig. 5a,b).

With backpropagation and aligned gain sensors, mean test accuracy increased from 0.6096 at D1 to 0.6752 at D2 and 0.9182 at D3. The paired D3-minus-D1 effect was 30.86 percentage points (95% seed-bootstrap interval 30.45–31.30; 10/10 positive pairs; exact sign-flip $P = 0.002$; Fig. 5c,d). The operating point was selected transparently by exploratory signal, coupling and gain ladders and then fixed before these ten inference seeds (Methods and Supplementary Fig. S15); the 31-point contrast must therefore be read as a replicated effect at a calibrated operating point. Physical depth was flat when the sensors were independent (0.02 points) or trial shuffled (-0.01 points), and decreased under exact-resource reversal of fine and global tree placement (-0.43 points, -0.67 to -0.15). The corresponding aligned-minus-control interactions were 30.84, 30.87 and 31.28 points, respectively, all positive in 10/10 paired seeds. Thus the effect requires correspondence between task factors, branch-local sensors and their physical ordering; stage count alone is insufficient.

The nonlinearity also mattered at this operating point. Aligned raw-additive BP decreased from 0.5733 at D1 to 0.5383 at D2 and 0.5190 at D3, whereas the matched shunting series increased (Fig. 5f). This control supports a divisive hierarchical-composition mechanism in this task, not a general impossibility for additive or expanded point networks.

A same-task architecture ladder sharpened that boundary (Fig. 5g–i). An all-active grouped-star model retained every branch module, mask, parameter, contact and persistent state but removed serial child-to-parent composition. It matched D1 (-0.005 points), remained near 0.610 at every depth, and therefore fell behind the serial tree by 6.54 points at D2 and 30.81 points at D3 (30.41–31.26; 10/10 positive). Reversing fine and global placement reversed the D3 contrast to -0.43 points, giving a 31.24-point alignment interaction (30.65–31.77; 10/10 positive). Thus the large effect requires serial composition in the model, rather than merely possessing the same collection of branch units. It is not an unconstrained capacity advantage: active- and total-parameter-matched point MLPs reached 0.9861 and 0.9902 and exceeded the serial D3 tree by 6.79 and 7.20 points, respectively, in every paired seed (Fig. 5l). The tree supplies a task-aligned structured resource, not a computation that a flexible point network cannot learn.

Local learning retained the aligned depth benefit. A single decoder-derived soma coordinate rose from 0.6083 at D1 to 0.7778 at D3 (16.95 points, 16.37–17.44; 10/10 positive), while exact path transport reached 0.8873 at D3 (27.90 points, 27.24–28.53; 10/10 positive; Fig. 5e). Exact-resource reversal removed both effects: the aligned-minus-reversed depth interactions were 17.72 points (17.27–18.18; 10/10 positive) for the shared coordinate and 28.68 points (28.07–29.27; 10/10 positive) for path transport. A separate replay on the 30 aligned BP checkpoints reproduced autograd with exact path transport (minimum gradient cosine 0.9999999999994), whereas mean shared-soma cosine fell from 0.946 at D1 to 0.767 at D3. The local-rule performance and checkpoint diagnostic together show that physical hierarchy can remain accessible to local eligibility when a sufficiently specific error is transported; they do not establish that exact path transport is itself biologically available.

An autograd teaching-coordinate ladder localized the remaining separation (Fig. 5j,k). Replacing each compartment's loss derivative by its owning soma derivative while retaining autograd eligibility and the BP optimizer cost only 0.64 points at D3 (0.15–1.17; 8/10 positive; $P = 0.033$). Because this arm and LocalCA initially used different optimizer groups, a transparently post-outcome diagnostic repeated soma broadcast with the frozen LocalCA optimizer. It then matched shared-soma LocalCA (-0.06 points, -0.23 to 0.09; $P = 0.492$), whereas changing only the broadcast arm's optimizer accounted for 13.47 points (12.89–14.05). Within LocalCA, path transport recovered 10.96 points over the shared coordinate (10.34–11.58); full BP retained a 3.09-point advantage over path LocalCA (2.48–3.79). In this calibrated task, the dominant BP–local gaps are therefore optimizer regime and within-tree coordinate specificity, not the tested local eligibility expression itself.

An alignment dose response tested whether the depth effect changed before the fully matched endpoint (Fig. 5m–o). Intermediate sensor alignments $\alpha \in \{0.25, 0.50, 0.75\}$, with $\alpha = 0$ denoting independent sensors and $\alpha = 1$ matched sensors, were registered after the $\alpha = 0$ and 1 outcomes were known, so this is a prospective interpolation rather than an independent confirmation. The D3-minus-D1 effect increased from 0.02 points at $\alpha = 0$ to 0.21, 0.82 and 3.13 points at the intermediate levels, before rising to 30.86 points at full alignment. Every seed had a positive within-seed linear slope (mean 25.84 points per unit α, 95% interval 25.43–26.26; exact sign-flip $P = 0.002$), but the curve was strongly nonlinear: the $\alpha = 0.75$ minus 0.25 increase was only 2.92 points (2.65–3.17), whereas the final 1.00 minus 0.75 increment was 27.73 points (27.35–28.11). The result therefore localizes a steep alignment boundary near the fully matched task, rather than supporting a generic linear dose law.

We then tightened the point control and changed task hierarchy (Fig. 5p–u). The earlier grouped-star control retained serial-shaped child aggregators as independent arms. A literal grouped-point emulation instead replaced each aggregator by a parameter-matched direct projection to the soma: every nonsomatic level kept its local branch module, mask and shunting computation, but received the external streams independently and never inherited another branch state. The replacement copied the corresponding initialized raw conductances, making D1 an exact initialization control. On the existing $H = 3$ task, grouped-point accuracy remained near 0.610 across D1–D3 (aligned D3-minus-D1, -0.01 points, -0.06 to 0.04), while serial-minus-grouped-point reached 30.87 points at aligned D3 (30.48–31.31; 10/10 positive; exact sign-flip $P = 0.002$). The same contrast was -0.44 points after placement reversal (-0.69 to -0.16), giving a 31.31-point alignment interaction (30.73–31.84; 10/10 positive). The earlier grouped star and the literal grouped point differed by only 0.06 points at aligned D3 (0.01–0.13; 7/10 positive; $P = 0.084$), so the conclusion does not depend detectably on which nonserial parameter-matched readout is used.

An independent $H = 2$ hierarchy then changed the sensor partition to two 32-feature tiers, the fixed branch inventory to 4/4 and the serial morphologies to D1 [8] and D2 $[4, 1]$, while retaining the calibrated task family and all other optimizer and stopping settings. Ten fresh paired seeds 10300–10309 were used. Serial BP increased from 0.6588 to 0.9672, a 30.84-point D2-minus-D1 effect (30.46–31.23; 10/10 positive; $P = 0.002$). Reversing the two sensor tiers changed that effect to -1.56 points, yielding a 32.40-point depth-by-placement interaction (32.10–32.70; 10/10 positive). The literal grouped point stayed flat in both placements (aligned, 0.02 points, -0.02 to 0.06; reversed, -0.02 points, -0.06 to 0.02), and serial BP exceeded it at aligned D2 by 30.82 points (30.41–31.23), with a 32.36-point architecture-by-placement interaction. Shared-soma LocalCA retained a 21.20-point aligned depth effect (19.34–23.00) and exact-path LocalCA retained 31.38 points (30.95–31.83); placement reversal made both effects negative, producing 22.93- and 33.36-point interactions, respectively. Every positive interaction was ordered in 10/10 paired seeds. This is an independent hierarchy-depth replication within the same synthetic mechanism-matched task family, not a second dataset or evidence that natural tasks generally favor serial trees.

Together, the new same-task controls separate four explanations that the original depth comparison could not distinguish. The parameter-matched point networks reject a dendrite-exclusive expressivity claim. The serial-minus-star and serial-minus-literal-point interactions identify ordered divisive composition, rather than branch count, parameter count or a particular nonserial readout, as the source of the aligned forward advantage; the $H = 2$ cohort shows that this conclusion is not unique to the original three-stage inventory. The optimizer-matched soma-broadcast comparison finds no detectable penalty from the tested local eligibility expression, whereas path-specific transport recovers 10.96 percentage points over a shared neuronal coordinate; the principal local-credit limitation in this task is therefore the teaching coordinate and its optimization regime. Finally, the alignment interpolation shows that the forward depth advantage appears only near a strongly matched task–sensor hierarchy. The experiments consequently support a conditional role for dendritic structure: it provides a useful structured computation and address system when the task matches the tree, but neither depth alone nor dendritic material guarantees better learning.

Reconstructed arbors supply sparse addresses whose gain conductance regulates

The evidence so far comes from synthetic trees with hand-specified topology. We next asked whether reconstructed cortical arbors provide such addresses. We constructed ancestry dictionaries from eight layer 2–5 intratelencephalic neurons in MICrONS materialization 1822, mapping 28,669 classified inputs to collapsed cable segments from one

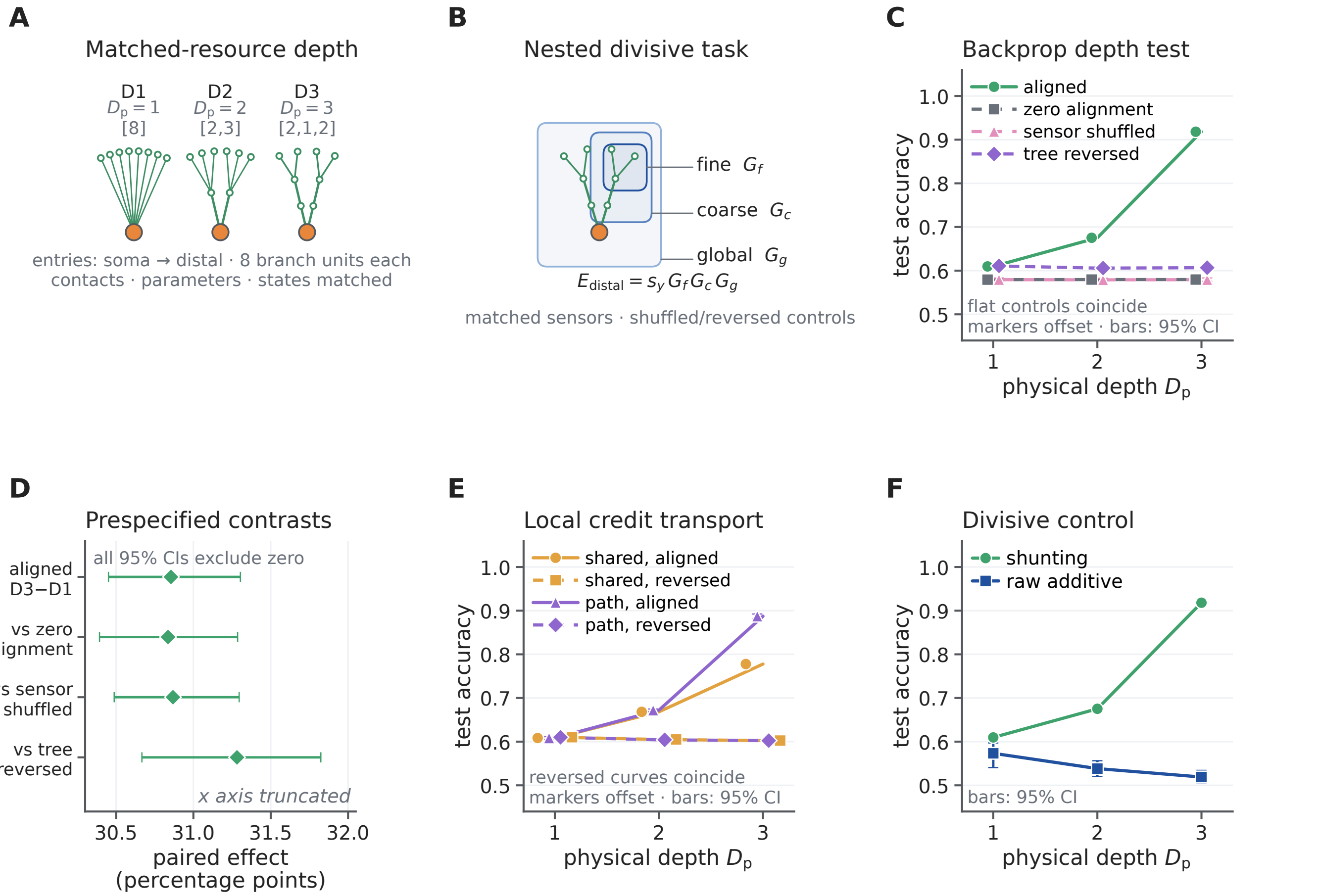

Figure 5: **Matched nonlinear hierarchy makes exact-resource physical dendritic depth useful.** **A**, D1–D3 have $D_p = 1, 2, 3$ stages. Bracket entries are soma-to-distal branching factors: [8], [2,3] and [2,1,2] give 8, $2+6$ and $2+2+4$ nonsomatic branch units, with identical contact, candidate-slot, parameter and state budgets. **B**, A distal positive-rate signal is multiplied by fine, coarse and global gains; independent sensors, trial-shuffled sensors and reversed physical placement are matched controls. **C**, Test accuracy under shunting backpropagation. Lines are ten-seed means and error bars are the 95% seed-bootstrap interval; coincident flat-control markers are offset. **D**, Prespecified paired D3-minus-D1 effect and aligned-minus-control interactions; bars are 95% paired-seed bootstrap intervals. **E**, Aligned and reversed-placement local credit assignment (LocalCA) with one shared-soma coordinate or exact path transport; the coincident reversed curves are drawn with offset markers. **F**, Aligned shunting and raw-additive BP. All inferential runs use fresh seeds 10200–10209.

mouse. Electrical parameters were normalized passive proxies, and reciprocal axial coupling was solved globally. The analysis complements prior connectomic proposals for anatomical credit assignment and the larger MICrONS morphology map [29, 30].

For neuron u, we define a K-channel feedback dictionary $\Phi_{u,K} \in \mathbb{R}^{N_u \times K}$, with rows for its synaptic or compartment locations and columns for available feedback fields. An *address* is one such column: its support and gain specify which locations can share one task-dependent coefficient. A point neuron has only the shared column $\mathbf{1}$; depth bins supply coarse columns; an ancestry dictionary supplies nested subtree columns; and an unrestricted dense rank-K subspace is a descriptive upper bound. This definition separates neuronal ownership from within-tree address and makes clear that dendrites supply structured, not arbitrary, credit bandwidth.

This distinction can be made quantitative. For a fixed family of K-column dictionaries $\Phi_{u,K}$ and a distribution of exact task-dependent coefficient fields $\boldsymbol{c}_u$, let $P_{\Phi_{u,K}}$ be the W-orthogonal projector onto $\mathrm{col}(\Phi_{u,K})$. For tolerated

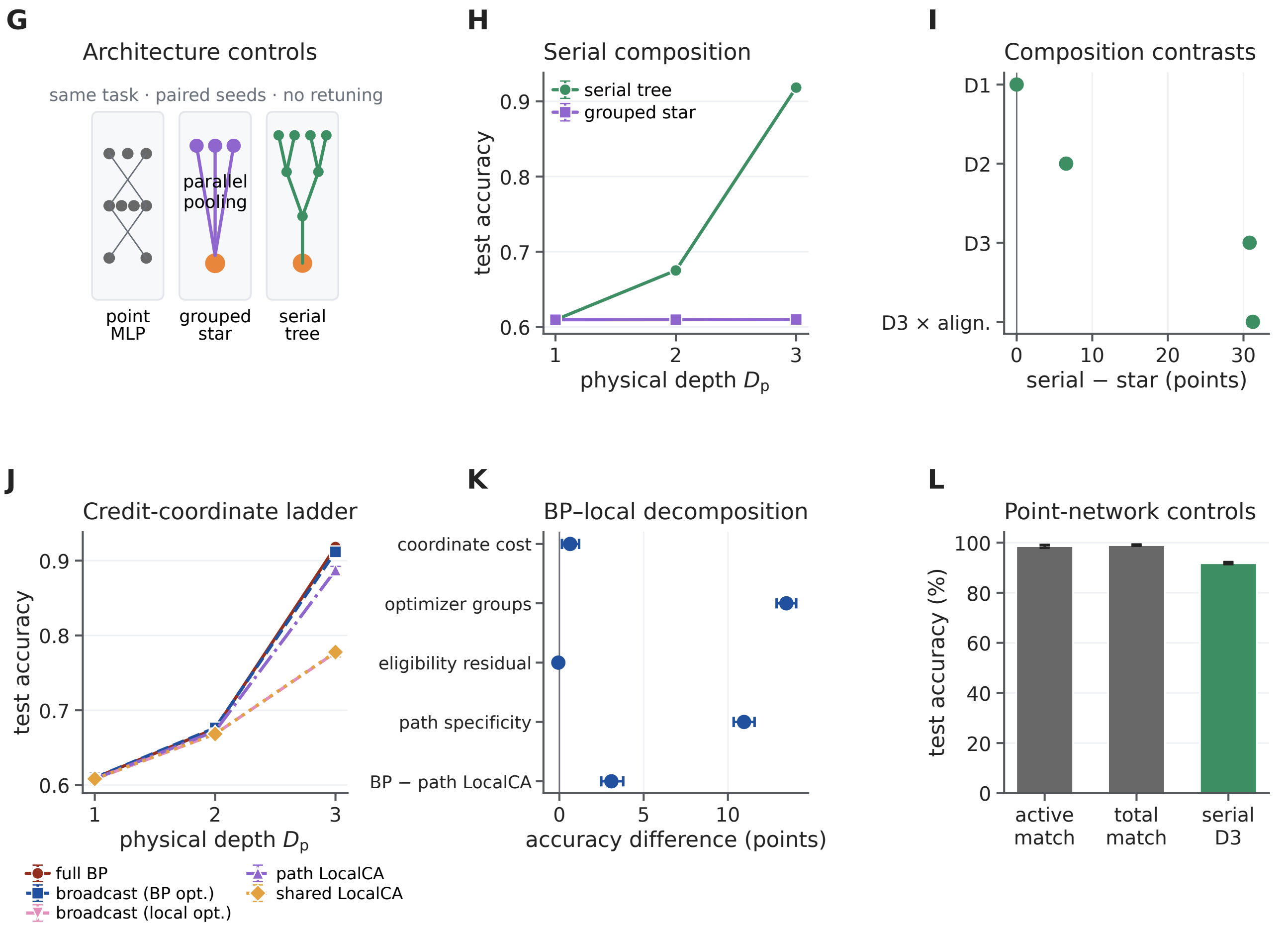

Figure 5: **Point–dendrite and BP–local-credit controls. G**, Same-task architecture ladder. **H**, Serial tree and exact-resource grouped-star accuracy across physical depth. **I**, Paired serial-minus-star contrasts and the D3 alignment interaction. **J**, Full BP, soma-broadcast autograd under the BP or LocalCA optimizer, and shared- or path-transport LocalCA. **K**, Paired D3 decomposition: teaching-coordinate cost and the optimizer contrast use soma-broadcast autograd; the eligibility residual compares optimizer-matched broadcast with shared LocalCA; path specificity compares the two LocalCA transports. **L**, Point MLPs matched to active or total parameters versus the serial D3 tree. Means and paired contrasts use the same ten seeds 10200–10209; error bars are 95% seed-bootstrap intervals. The optimizer-matched amendment is post-outcome and diagnostic.

unexplained field-energy fraction $0 < \varepsilon < 1$, define the ε-address dimension

$$d_{\varepsilon,u} = \min\left\{ K : \frac{\mathbb{E}\|P_{\Phi_{u,K}} \boldsymbol{c}_u\|_W^2}{\mathbb{E}\|\boldsymbol{c}_u\|_W^2} \geq 1 - \varepsilon \right\}. \tag{10}$$

A point neuron with one neuronal teaching coefficient has rank-one task-dependent address space, irrespective of its number of synapses. A K-route dendritic dictionary has address rank at most $\operatorname{rank}(\Phi_{u,K}) \leq K$ and can express K structured coefficient fields without assigning those routes to K different neurons. This is a resource statement, not a claim that dendrites create arbitrary feedback bandwidth: a point-neuron system can emulate the same field if it is supplied with K separately routed teaching coefficients or an equivalent gated feature expansion (Supplementary Information).

For an excitatory site i and candidate inhibitory route k, the direct topological route matrix was

$$\Gamma_{ik} = A_{ik}\beta_k, \tag{11}$$

where A_{ik} marks route ancestry and $\beta_k = g_k^I / g_k^{\text{tot}}$ is normalized inhibitory gain. This is a route-sensitivity proxy, not

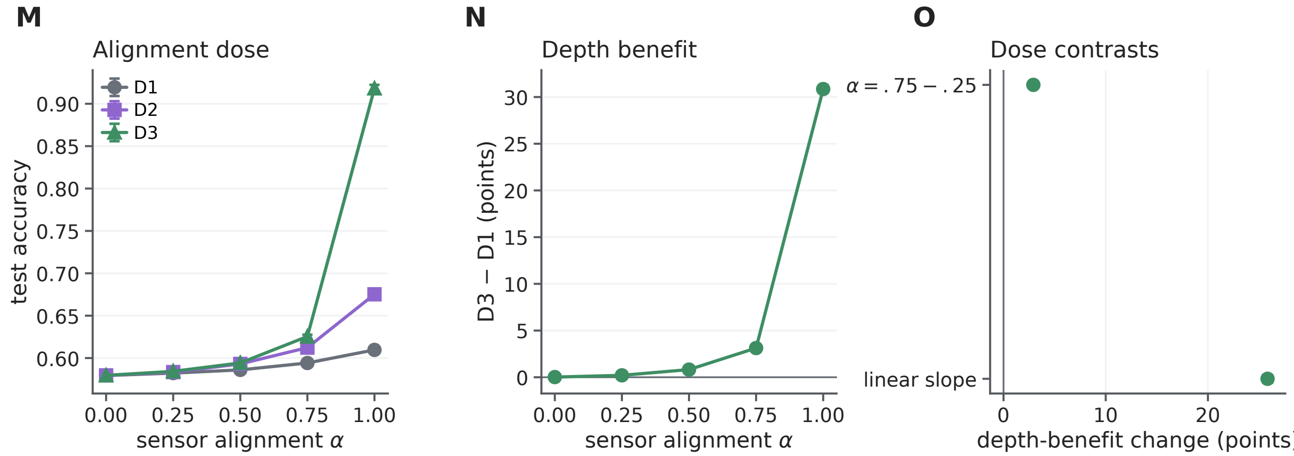

Figure 5: **Physical-depth benefit has a steep task–sensor alignment boundary.** **M**, Test accuracy across sensor alignment α and physical depth. **N**, Paired D3-minus-D1 effect. **O**, Change in that depth effect between $\alpha = 0.25$ and 0.75, and the prespecified within-seed linear slope across all five alignment levels. Lines are ten-seed means; all error bars are 95% paired-seed bootstrap intervals. The intermediate alignment levels were registered after the two endpoints were known and constitute a prospective interpolation, not an independent task replication.

the exact passive-cable derivative used below. Its nested domains addressed a median 2.67% of mapped excitatory input and had mean participation rank 20.81, versus rank one for a scalar broadcast (Fig. 6a–c).

We measured the fraction of a modeled field retained by K feedback channels. For weighted field $\overline{\boldsymbol{g}} = W^{1/2}\boldsymbol{g}$ and dictionary $\overline{\Phi}_{T,K} = W^{1/2}\Phi_{T,K}$, let $\overline{P}_{\Phi}$ be the Euclidean projector onto $\mathrm{col}(\overline{\Phi}_{T,K})$. Field capture is

$$\mathscr{C}_K(\boldsymbol{g};\Phi,W) = \frac{\|\overline{P}_{\Phi}\overline{\boldsymbol{g}}\|_2^2}{\|\overline{\boldsymbol{g}}\|_2^2} = 1 - \frac{\|\overline{\boldsymbol{g}} - \overline{P}_{\Phi}\overline{\boldsymbol{g}}\|_2^2}{\|\overline{\boldsymbol{g}}\|_2^2}. \tag{12}$$

Here W contains summed mapped excitatory area, so Eq. 12 is first an anatomical field-energy measure. Only if $\overline{\boldsymbol{g}}$ is an exact gradient in explicitly transformed coordinates $\overline{\boldsymbol{w}} = W^{-1/2}\boldsymbol{w}$ and the transformed loss has L_W-Lipschitz gradients does $\mathscr{C}_K$ also equal the ratio of projected and full one-step descent guarantees at step $1/L_W$ (Supplementary Information).

We led with an independent generator: exact state-matched responses to focal shunts in the reciprocal cable model. At eight channels, ancestry routes captured 0.458, versus 0.363 for random paths and 0.249 after row shuffling, with both advantages positive in all eight cells. Differences from depth bins (0.014) and degree- and depth-matched surrogate trees (0.035) were inconclusive: the cell-bootstrap interval for the latter excluded zero, whereas the primary paired Wilcoxon test did not ($P = 0.078$). We therefore infer sparse support beyond random assignment but not a demonstrated advantage of fine ancestry beyond coarse depth and degree (Fig. 6i,j).

As a model-matched ceiling, we then generated fields directly from K.

At eight channels, morphology-selected routes captured 0.487 of the energy in the model-derived route-perturbation fields using 6.92% of the connections in a dense feedback matrix. This was 85.0% of the field energy captured by an unconstrained dense eight-channel oracle. Random paths captured 0.285, depth bins 0.222, and ancestry-shuffled routes 0.198. The morphology advantage was positive in every cell for each control; hierarchical cell-and-stream intervals excluded zero (Fig. 6d–g). Normalizing by wiring makes the resource statement explicit: per unit of feedback wiring, morphology-selected routes captured 14.2 times more field energy than the dense oracle (95% cell-bootstrap interval 10.7–18.2), whereas the ancestry-shuffled control, matched at the identical 6.9% wiring density, reached 5.3-fold; the anatomy-specific advantage at matched density is therefore 2.7-fold, with the remainder contributed by sparsity itself (Fig. 6k,l). The ordering also survived an analysis restricted to directly typed presynaptic E/I inputs: morphology captured 0.696, compared with 0.347 for random paths, 0.447 for depth bins, and 0.308 after ancestry shuffling (Fig. 6h).

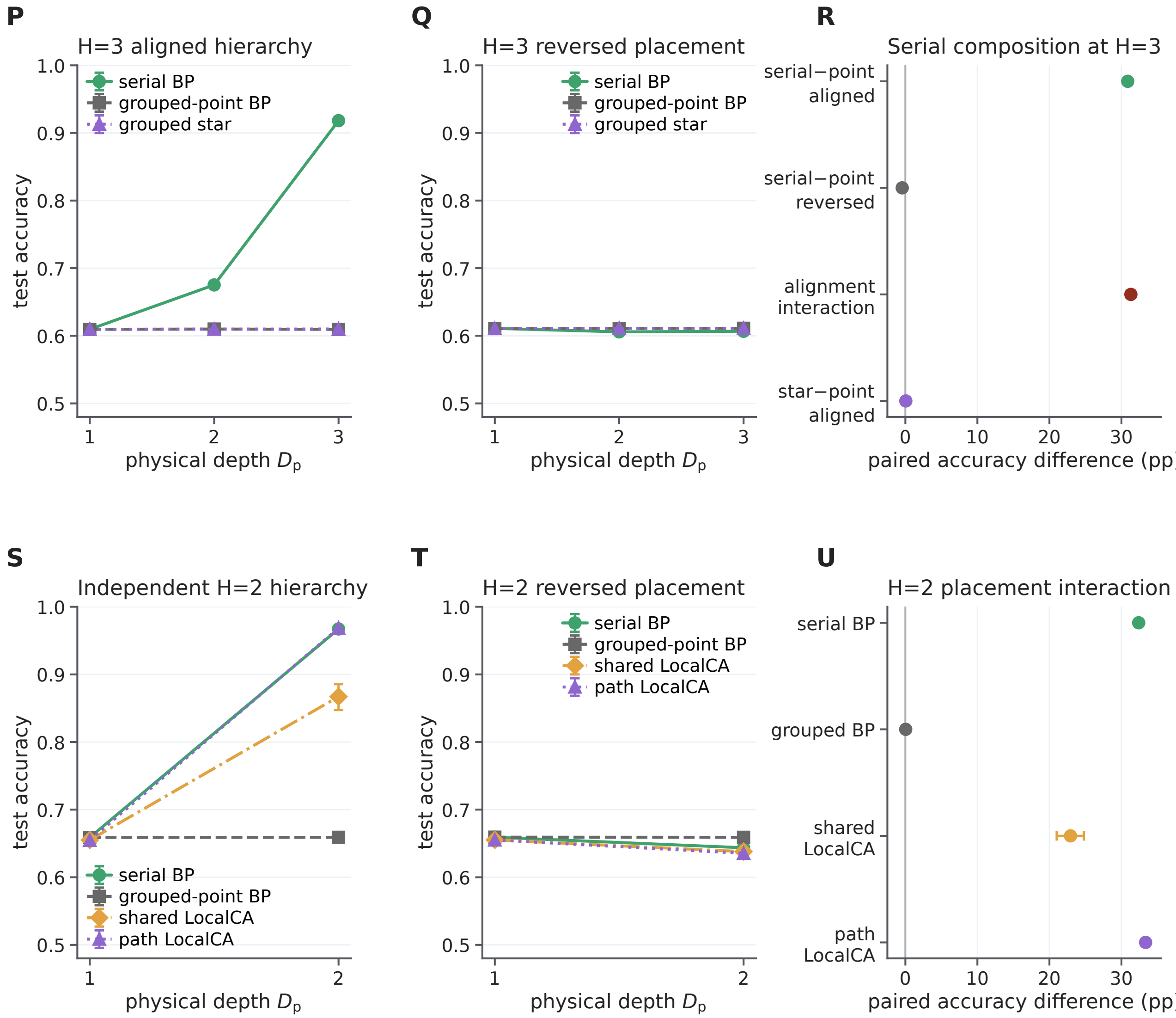

Figure 5: **Serial composition replicates across two task hierarchies and vanishes in literal grouped-point controls. P,Q**, Existing $H = 3$ serial tree, literal grouped-point emulation and earlier grouped star under aligned and reversed placement. **R**, Paired H3 serial-minus-point contrasts, their alignment interaction and the grouped-star-minus-point comparison. **S,T**, Fresh $H = 2$ serial BP, grouped-point BP and serial LocalCA with a shared-soma or exact-path coordinate under aligned and reversed placement. **U**, Paired aligned-minus-reversed D2-minus-D1 interactions. Lines are means across ten paired seeds; all error bars are 95% seed-bootstrap intervals. The $H = 2$ cohort changes hierarchy and inventory within the same calibrated synthetic task family and is not an independent dataset.

In 47 additional cells, morphology captured 0.756 with 8.08% of dense wiring, versus 0.302–0.441 for structural controls; every within-cell contrast was positive (Supplementary Fig. S10). The cohort is nucleus-disjoint but from the same mouse and a convenience-derived historical reconstruction.

All comparisons matched channel count. They establish anatomical compression of modeled fields, not a task gradient or route use by the animal.

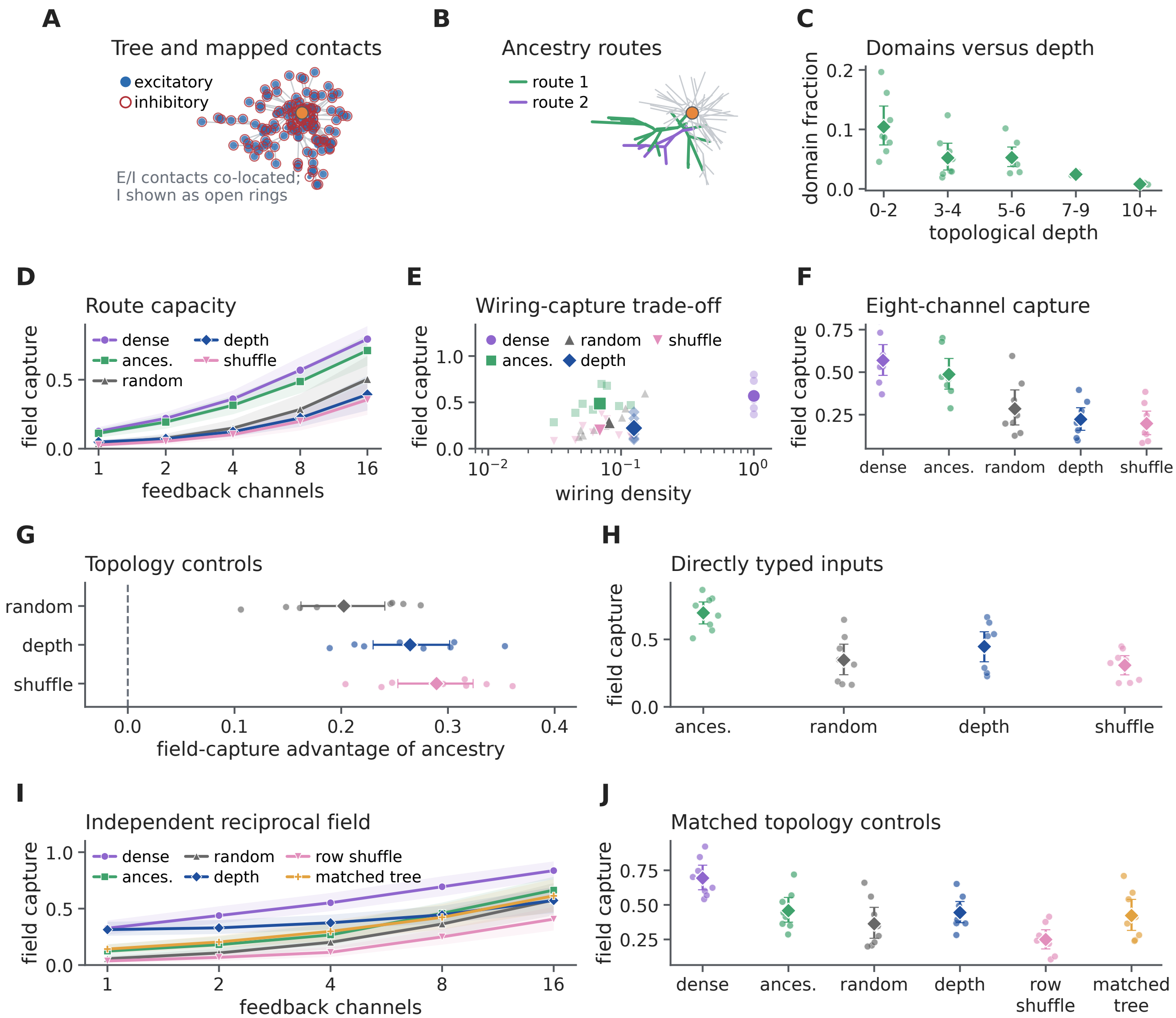

Figure 6: **Reconstructed arbors provide a sparse, mainly coarse routing basis for modeled perturbations. A**, Principal-plane projection of an example reconstructed tree (root 864691135060651419) with mapped excitatory and inhibitory contacts. **B**, The same tree with two illustrative nested ancestry-defined route domains. **C**, Credit-domain fraction across topological-depth bins for excitatory-bearing segments, first averaged within cell and depth bin; small points are cell means and diamonds are means with 95% cell-bootstrap intervals. **D**, Weighted field-energy capture as a function of feedback-channel count for a dense principal-component oracle, morphology-selected paths, random paths, depth bins and ancestry-shuffled paths. Lines are means across eight cells; shading is the cell-bootstrap 95% interval after averaging Monte Carlo streams. **E**, Eight-channel weighted field-energy capture versus feedback wiring density. Small points are cell means and large markers are means across cells. **F**, Absolute eight-channel capture for all five dictionaries. **G**, Within-cell morphology advantage over each structural control. **H**, Eight-channel weighted field-energy capture after restricting the analysis to directly typed presynaptic inputs. Because this subset has a different E/I composition and effective rank, it is a sensitivity analysis and is not compared numerically with the all-input capture in **F**. **I**, Capture of an independent field generated by exact focal-shunt responses of the reciprocal cable model. **J**, Eight-channel capture of that field. The matched-tree control preserves node depths and parent degrees; row shuffling preserves selected column values. Coordinates are weighted by summed mapped excitatory synaptic area. All fields are modeled responses on measured anatomy, not observed task gradients. Points in **F–H,J** are cell means; diamonds and intervals summarize the eight cells.

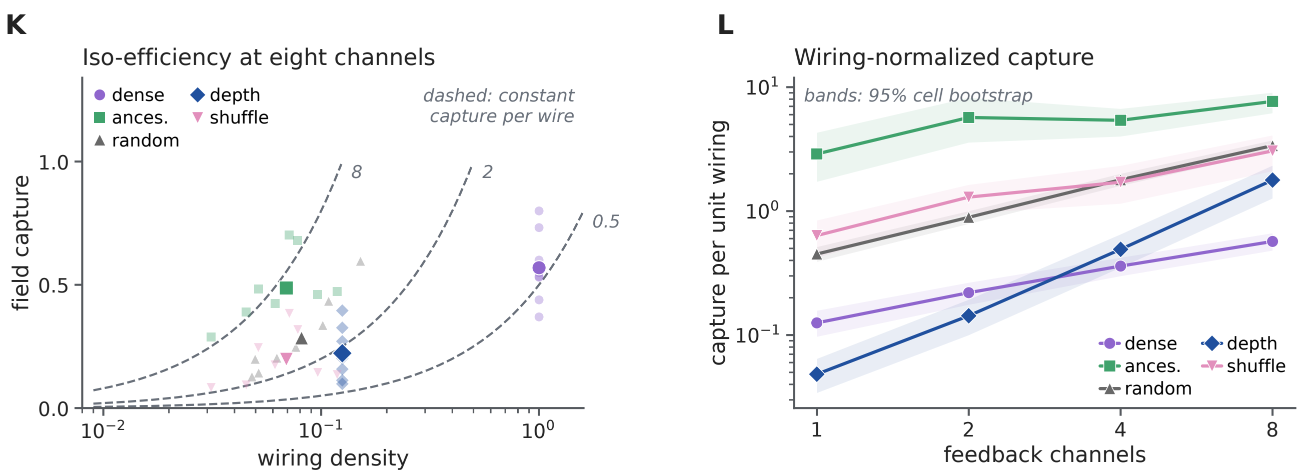

Figure 6: **Sparse route dictionaries trade feedback wiring for field capture.** **K**, The eight-channel capacity comparison replotted on a logarithmic wiring axis. Dashed curves mark constant capture per unit wiring; small points are reconstructed-cell means and large markers are means across the eight cells. **L**, Capture per unit wiring across channel budgets. Lines are cell means and bands are 95% cell-bootstrap intervals. At eight channels, ancestry routes provide a model-matched 14.2-fold advantage over the dense oracle, while the density-matched ancestry shuffle reaches 5.3-fold; their ratio gives the 2.7-fold anatomy-specific component. Wiring counts feedback connections, not cable length or metabolic cost.

The route analysis above used inferred inhibitory distributions. We next asked whether synapse-resolved inhibitory connectivity is organized in a manner compatible with branch-level control. This question builds on physiological and modeling work on tunable dendritic inhibition and inhibitory plasticity [31, 32]. We joined the disjoint reconstruction cohort to the published MICrONS inhibitory census [33]. Twenty of 47 target cells could be joined by stable nucleus identity across releases; five had different root IDs after re-segmentation. The source tables contained 116,401 input locations and 2,789 known contacts from 116 inhibitory cells. Of these, 114,775 inputs and 2,637 contacts from 114 inhibitory cells mapped within 5 μm. This is a structural analysis of one mouse; it does not identify a learning signal, and the packaged census does not provide a per-target proofreading-status field.

For each inhibitory cell–target connection, we retained one representative contact per independent axonal clump. The 402 connections with at least two clumps provided a within-axon test. Repeated contacts from the same inhibitory axon were 12.0 μm closer along the target tree than compartment- and path-distance-matched input locations (95% bootstrap interval -21.5 to -4.2 μm; 15 of 20 target means were negative; one-sided Wilcoxon $p = 0.0032$). They also shared 0.0357 more normalized path (0.0091–0.0766; 15 of 20; $p = 0.0028$) and addressed descendant input domains with 0.0322 greater weighted cosine overlap (0.0060–0.0743; 14 of 20; $p = 0.0053$; Supplementary Fig. S12d–f). These target-level comparisons do not by themselves separate dendritic organization from repeated sampling of the same axons or local tissue geometry. Averaging first within each of 92 presynaptic axons retained the tree-distance effect ($p = 0.0008$), whereas the shared-path and domain-overlap effects were not reliable ($p = 0.104$ and $p = 0.393$). With candidate locations jointly matched for path distance and three-dimensional proximity, none of the three contrasts was reliable ($p = 0.108$, 0.826, and 0.54, respectively). Thus repeated contacts are spatially co-targeted, but these data do not establish geometry-independent organization over descendant credit domains.

Inhibitory classes exposed different spatial scales. After retaining one contact per axonal clump, distal dendrite-targeting contacts addressed a smaller fraction of target-cell input than perisomatic-targeting contacts in all 20 cells. The mean within-target difference of the class medians was -0.0162 (-0.0231 to -0.0105; one-sided $p = 9.5 \times 10^{-7}$; Supplementary Fig. S12g). This class effect should not be confused with a global placement advantage. Across all inhibitory contacts, descendant-domain size did not differ reliably from matched generic input locations (mean difference 0.0016, $p = 0.546$; Supplementary Fig. S12h).

Finally, we measured the internal compressibility of the actual inhibitory route dictionary. At eight sites, its highest-leverage columns captured 0.743 of that same weighted dictionary, close to a dense singular-vector oracle (0.774), compared with 0.379 for random actual sites, 0.397 for depth bins, and 0.369 after depth-preserving ancestry shuffling. Each selected-route contrast was positive in all 20 targets (one-sided $p = 9.5 \times 10^{-7}$ for each; Supplementary Fig. S12i,j). Selection by leverage is an anatomical compression oracle: it measures redundancy within a dictionary using columns of that dictionary, not independent credit routing or a claim that the animal chose those sites. Together, the census establishes class-dependent spatial scales and available nested domains, while the axon-clustered and 3D-matched controls prevent a stronger functional-routing interpretation.

Topology-based reconstruction does not show that conductance can selectively control the routes. At the textbook passive calibration ($R_m = 15{,}000\,\Omega\,\mathrm{cm}^2$), the primary contrast was effectively null (-0.0019, two of eight cells positive); localization emerged only in permissive normalized or higher-conductance regimes. We therefore specified a focal intervention before examining its outcome. This adjoint-credit question extends prior analyses of spatial inhibitory domains in dendrites [4]. The passive conductance matrix of each reconstructed tree was denoted by G, with voltage $\boldsymbol{V} = G^{-1}\boldsymbol{b}$, where $\boldsymbol{b}$ is the current-source vector. Let $\boldsymbol{e}_s$ and $\boldsymbol{e}_k$ be standard basis vectors at the soma and focal compartment k, respectively. For a somatic loss, the adjoint error is $\boldsymbol{q} = G^{-1}\boldsymbol{e}_s\,\partial\mathscr{L}/\partial V_s$. A shunt of conductance κ_k at compartment k changes the transport operator according to

$$\boldsymbol{q}' = \boldsymbol{q} - \frac{\kappa_k q_k}{1+\kappa_k (G^{-1})_{kk}} G^{-1}\boldsymbol{e}_k. \tag{13}$$

Writing $\boldsymbol{c}_k = G^{-1}\boldsymbol{e}_k$ and $r_k = (G^{-1})_{kk}$ gives $\Delta\boldsymbol{q} = -a_k\boldsymbol{c}_k$, where $|a_k| = \kappa_k|q_k|/(1+\kappa_k r_k)$. Thus dose controls the response amplitude, which increases monotonically and saturates at $|q_k|/r_k$, whereas the cable column $\boldsymbol{c}_k$ controls its spatial spread. For descendant set $D(k)$ and a matched comparison set $C(k)$, the transport-only selectivity criterion is

$$S_k = \frac{\mathrm{median}_{i\in D(k)}\,|c_{k,i}|}{\mathrm{median}_{j\in C(k)}\,|c_{k,j}|} > 1. \tag{14}$$

Equation 14 separates electrotonic selectivity from perturbation strength. It is necessary for descendant enrichment of absolute adjoint change under this median contrast, but is not sufficient for the synaptic-gradient statistic, which also depends on baseline adjoints and local driving forces. A first-order-current-matched additive perturbation changes the voltage-driving term but not G^{-1}. This yields a direct test: after a state-matching compensatory current injection restores somatic voltage and the scalar output error, conductance shunting should alter exact synaptic gradients more strongly among descendants than among depth-matched unrelated sites. Reciprocal coupling still changes credit outside the descendant set.

We tested 101 depth-stratified focal sites across the eight cells; the localization profile across topological depth is shown in Fig. 7e. At the unit dose, the descendant-minus-depth-matched-unrelated localization index—descendant minus depth-matched unrelated median log gradient attenuation—was 0.104 for shunting and 0.035 for the matched additive perturbation. The cell-level paired difference was 0.069 log units (95% bootstrap interval 0.053–0.086), positive in all eight cells. The contrast rose monotonically with dose, from 0.018 at 0.25 local-conductance units to 0.128 at two units. True focal relations also exceeded relations sampled from other focal sites, stratified by topological-depth quartile when another site was available, by 0.080 (0.059–0.098), again in all eight cells (Fig. 7b–d).

Because a synaptic gradient is the product of adjoint error and local driving force, we next froze either factor algebraically. Keeping the full post-shunt voltage and driving force but replacing the post-shunt adjoint with its baseline value gave localization 0.026. Restoring the post-shunt adjoint raised it to 0.104, an exact transport-specific difference of 0.079 that was positive in all eight cells (Fig. 7h). This comparison holds the post-shunt driving-force field fixed and therefore isolates the contribution of changed error transport without relying on first-order current matching. The complementary adjoint-only state gave localization 0.130. A two-factor Shapley sensitivity analysis assigned 0.087 of the planned shunt-minus-additive contrast to the adjoint change and -0.017 to the driving-force change, reconciling exactly to the 0.069 net contrast at every site (Supplementary Section S8). These factor substitutions are algebraic controls, not separate biological interventions.

The state-matching residual at the soma was numerically zero, conductance matrices remained positive definite, and adjoint gradients agreed with finite differences to a maximum relative error of 1.03×10^{-6}. The principal contrast also survived changes in normalized conductance scale and inhibitory reversal potential (Fig. 7f). Restricting both synaptic weighting and focal sites to directly typed inputs gave a shunt-minus-additive contrast of 0.101 (0.075–0.129), positive in all eight cells (Fig. 7g). In the disjoint sensitivity cohort, the same contrast was 0.138 (0.116–0.162) across 235 sites in 45 eligible cells and was positive in every cell (Supplementary Fig. S10f). These results identify a conductance-specific gradient-localization mechanism at fixed somatic state in the normalized proxy model.

We next calibrated axial and leak conductances from reconstructed radii and lengths using explicit specific resistances. With $R_a = 150\,\Omega\,\mathrm{cm}$ and $R_m = 15{,}000\,\Omega\,\mathrm{cm}^2$, the median axial-to-leak ratio was 68.0. The shunt-minus-additive contrast was then -0.0019 in the original cohort (two of eight cells positive) and 0.0024 in the v661 cohort (45 of 45 positive). Lower effective membrane resistance, representing a progressively higher-conductance state, restored a larger effect: at $R_m = 1{,}000\,\Omega\,\mathrm{cm}^2$ the original eight-cell contrast was 0.0164, and at $R_m = 300\,\Omega\,\mathrm{cm}^2$ it was 0.0674 (eight of eight positive in both conditions; Fig. 7i). Low input resistance and conductance-driven gain modulation are established features of in-vivo-like high-conductance states [34, 35]. These values use an explicit relative calibration of synapse size because MICrONS synaptic area is not a conductance measurement. The focal signature is therefore an electrotonic-regime-dependent model prediction, not a universal property of the reconstructed arbors or evidence that inhibitory synapses carried endogenous teaching signals in the animal.

We then froze a passive sensitivity matrix that removes the relative-dose confound. Across the same 101 sites and eight cells, it crossed three membrane resistances, three levels of distributed leak-like background conductance, and relative-local, fixed-absolute and input-conductance-normalized doses (16,362 site–condition–intervention rows). At a fixed 0.5-nS shunt, the shunt-minus-additive localization contrast was 0.0104 at $R_m = 300\,\Omega\,\mathrm{cm}^2$, 0.0073 at $R_m = 1{,}000\,\Omega\,\mathrm{cm}^2$, and -0.0039 at $R_m = 15{,}000\,\Omega\,\mathrm{cm}^2$; the first two were positive in all eight cells, whereas only two of eight were positive in the last regime. At the high-resistance setting, adding one and four times the baseline leak as distributed background conductance raised the input-conductance-normalized contrast from -0.0004 to 0.1210 and 0.2587, respectively. Every tested descendant gradient was attenuated in magnitude; none was enhanced or changed sign. Thus passive shunting supplies graded attenuation, not signed reversal, and its conductance-specific spatial advantage requires both sufficient dose and a selective electrotonic column (Supplementary Fig. S11).

Finally, we asked whether the same localization survives weak active conductances rather than a purely passive operator. The ensemble was evaluated at the permissive electrotonic operating point identified above ($R_m = 1{,}000\,\Omega\,\mathrm{cm}^2$ with one baseline-leak unit of distributed background conductance; Methods); it extends the localization result within that regime rather than restoring it at the high-resistance calibration. For each of the eight cells, we solved 64 channel-density draws containing voltage- dependent Na, K, Ca, HCN and NMDA currents, then linearized the exact nonlinear steady-state residual to obtain its local Jacobian. All 512 equilibria were accepted; every Jacobian remained positive definite and the maximum equilibrium residual was 9.6×10^{-11}. Draws were nested sensitivities, not independent cells. At the unit input-conductance-normalized dose, focal shunting produced descendant localization 0.529 (95% cell-bootstrap interval 0.513–0.541), attenuated every modeled descendant gradient, produced no sign flips and reduced descendant gradient energy to 0.330 of baseline (Fig. 7j–m). Its localization exceeded the state-matched additive control by 0.382 (0.356–0.409), positive in all eight cells (two-sided Wilcoxon $P = 0.0078$). The effect rose monotonically across the three doses. This active steady-state ensemble strengthens the local-Jacobian prediction but is not a kinetic-channel or spiking simulation.

Measured responses reveal a boundary for morphology-specific alignment

Structural capacity used ancestry-kernel perturbations and the focal result used modeled adjoints. We next tested whether the routes align with measured functional inputs and response-derived task gradients. A MICrONS/DANDI join identified seven reconstructed targets and 136 repeated visual conditions per scan. The original one-scan-per-target cohort contained 69 connected, imaged excitatory partners. Functional similarity did not increase consistently with shared ancestry after controlling for Euclidean separation and depth: the mean target-level partial rank effect was -0.069 (95% target-bootstrap interval -0.255 to 0.108), positive in three of seven cells (Fig. 8a–c). This estimand asks whether within-tree position adds organization beyond geometry and complements the broader like-to-like wiring principle reported in the same MICrONS volume [36]. Using the deterministic rule "all automatic-conservative scans with at least five connected imaged partners" expanded the analysis to 13 scan observations. After averaging scans within target, the partial shared-path effect was -0.048 (-0.224 to 0.118), positive in three of seven cells. The scan-complete analysis therefore removed the selection ambiguity without changing the null conclusion.

We trained a branch surrogate for each postsynaptic response and projected its exact error into four-channel dictionaries. Each partner mapped through its dominant contact to one nonlinear state on the first major branch below the soma. Sites on that branch therefore shared one task-dependent error. With only 3–8 occupied major branches per target, this cannot resolve within-branch credit or strongly test fine topology. Stimulus identities were held out. Mean held-out field capture was 0.999 for a training-only dense oracle, 0.475 for morphology paths, 0.564 for nonempty random paths, 0.621 for depth bins, 0.446 for ancestry-shuffled paths, and 0.222 for a scalar broadcast (Fig. 8d). Ancestry routes minus ancestry-shuffled capture was 0.028 (95% interval -0.146 to 0.206), with three of seven target cells positive (Fig. 8f).

Learning outcomes gave the same boundary. Mean normalized held-out error was 0.778 for exact backpropagation, 0.781 for the dense oracle, 0.791 for depth bins, 0.801 for scalar broadcast, 0.834 for random paths, 0.837 for morphology paths, and 0.842 after ancestry shuffling. Thus morphology was close to its ancestry-shuffled control but did not outperform the simpler depth-bin or scalar dictionaries on this task. The paired morphology improvement was 0.004 normalized-error units (95% interval -0.021 to 0.027), positive in four of seven targets (Fig. 8e). A pooled capture–performance association vanished after centering within target ($\rho = -0.150$; label-permutation $P = 0.419$; Fig. 8g). Higher within-target capture predicted lower error at one channel ($\rho = -0.621$, $P = 0.0006$) and eight channels ($\rho = -0.480$, $P = 0.0104$), but not two or four (Source Data). Channel-wise morphology-minus-shuffle contrasts (Fig. 8h), reliability thresholds and a matched subset did not establish morphology-specific benefit.

We then replaced the coarse branch surrogate with an exact passive solve on each complete compressed reconstructed tree. Every imaged partner supplied a trainable non-negative conductance at its mapped segment; the exact local rule used the trial-specific soma-to-segment adjoint. Restricted rules used only scalar output error, local eligibility and a fixed four-channel feedback vector calibrated from topology before learning. All 13 eligible scans and ten held-out-stimulus splits per scan completed without exclusion, giving 520 method fits. Split replicates were averaged within scan, then scans within each of seven target cells.

Exact compartment error achieved mean normalized held-out MSE 0.788. The topology-matched, random-route and site-shuffled rules reached 0.832, 0.833 and 0.839, respectively (Fig. 8i). Topology improved MSE over site shuffling by 0.0070 units (95% target-bootstrap interval -0.0207 to 0.0321) and over random routes by 0.0013 (-0.0090 to 0.0113), neither a reliable advantage. At the common exact checkpoint, topology captured 0.232 of the eligibility-weighted update field, below site-shuffled routes (0.439) and random routes (0.358; Fig. 8j). Restricted dictionaries had identical effective rank within every scan–split run, and exact conductance gradients passed finite differences with maximum relative error 3.1×10^{-7}.

Thus route availability did not imply route use in either the coarse or segment-resolved task. Four dense dimensions nearly matched exact learning in the coarse model, but anatomical routes did not align reliably with measured- response credit on the full tree. The dense-oracle capture curve indicates why the coarse-model null is expected under the credit-operator account: a single channel already captured 0.965 of this response family's credit energy (effective rank 1.07), so a four-channel budget exceeds the task's credit rank and every equal-rank dictionary spans it—the regime in which the theory predicts the observed tie (Fig. 8o). Oracle calibration makes each restricted method a capacity ceiling. The null does not exclude other response windows, higher input coverage or electrophysiologically fitted conductances.

Measured responses showed no reliable morphology-specific alignment; the conditional claim, however, predicts that the same dictionary should become useful exactly when task credit rotates into its span. To test the conditional statement directly, we generated alignment-controlled quadratic objectives on each reconstructed tree. Exact-gradient fields were rotated from the span of the morphology dictionary into its orthogonal complement while channel count, dictionary, gradient energy, and loss curvature were held fixed. The same fields were evaluated with morphology, random-path, depth-bin and ancestry-shuffled dictionaries before norm-matched or iterative projected descent (Fig. 8k–n and Supplementary Information).

The morphology dictionary captured exactly the imposed aligned energy: from 0 at complete orthogonality to 1 at complete alignment. At 40% alignment it captured 0.400 and supported 0.615 of the loss reduction from a norm-matched full-gradient step. Its capture advantage over the three controls ranged from 0.203 to 0.228 and was positive in all eight cells. At complete orthogonality, morphology lost to every control in all eight cells; at complete alignment, it exceeded the controls by 0.710–0.799 and won in all eight. Thus the same fixed anatomy changes from an unhelpful to a useful feedback basis as only its alignment with task credit changes. Across the three control dictionaries and alignment levels, initial capture tracked 20-step progress within each cell (median Spearman $\rho = 0.990$, range 0.969–0.996; the one- and 20-step relations are shown in Fig. 8n and Supplementary Fig. S5).

This constructive sufficiency test imposes alignment, fits oracle projection coefficients and uses a quadratic objective. It links field capture to optimization under controlled conditions but does not show that MICrONS cells used these gradients; independently measured responses showed no reliable morphology-specific benefit.

The theory assumes that a neuron-specific teaching coordinate is available. We tested that assumption using published source data from a six-animal brain–computer-interface experiment in which the causal signs of two selected neurons, P+ and P-, were fixed by the experimental mapping [37] (Supplementary Fig. S17a). We defined one contrast before inspecting its sign: standardized dendritic population residual during error reduction minus the residual during error increase. If the dendritic signal follows causal credit rather than one common scalar error, the contrast should be positive around P+ and negative around P-.

The P+ contrast was positive in five of six animals (mean 0.0753; 95% bootstrap interval 0.0295–0.1214; exact one-sided random-sign permutation $P = 0.0313$). The P- contrast was negative in all six (mean -0.1498; -0.2054 to -0.0866; $P = 0.0156$). Their signed separation was positive in all six animals (mean 0.2251; 0.1416–0.3108; $P = 0.0156$; Supplementary Fig. S17b). Decomposing the six animal contrast vectors showed that 83.7% of pooled squared projection energy lay in the signed P+/P- mode, compared with 16.3% in a common scalar mode (animal-bootstrap 95% interval for signed-mode energy, 60.6–95.7%; leave-one-animal-out range, 76.5–90.2%)

(Supplementary Fig. S17c); descriptive neuron-level residual distributions from the published source table are shown in Supplementary Fig. S17d. With six animals, these exact permutation tests sit at or one step above their resolution floor. The analysis independently supports the first level of the theory—signed neuron identity—but does not determine how the signal is transported within a dendritic tree or uniquely implicate shunting.

Together, the experiments occupy a common alignment-by-bandwidth plane (Fig. 8o). Low bandwidth loses neuronal identity, whereas high bandwidth can make equal-rank spans coincide. Between these limits, ancestry routes help only when task credit falls sufficiently inside their span. The trained $K = 4$ interior optimum, the spectral and controlled- alignment sweeps, the credit-reversal task and both measured-response nulls all fall on the corresponding side of this boundary. The shaded regions are theory-derived regimes, not a fitted classifier of the experiments.

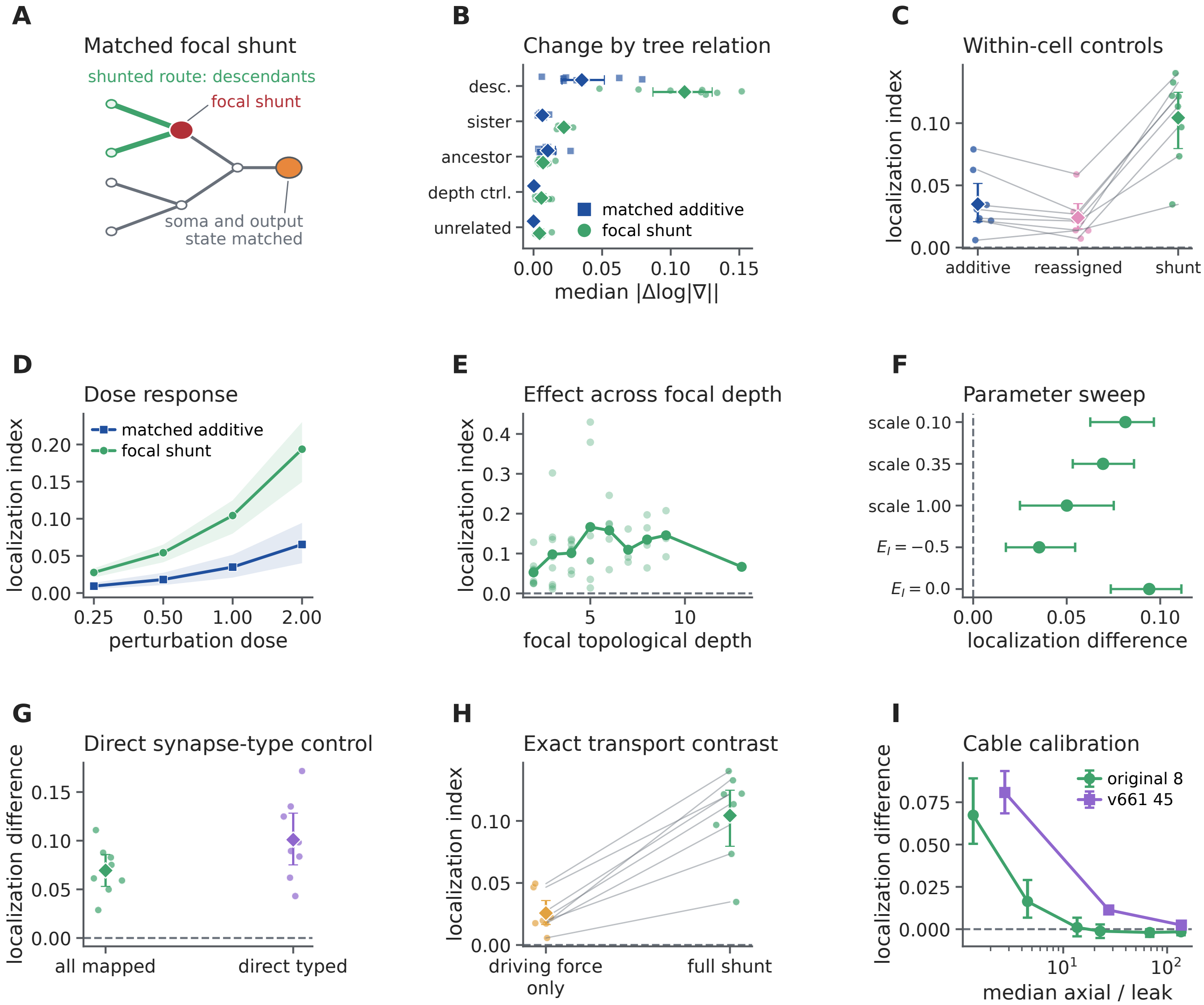

Figure 7: **Focal shunting produces a descendant-enriched modeled-credit signature in defined electrotonic regimes.** **A**, Focal intervention with state-matching somatic current. **B**, Absolute log-gradient change across descendants, sisters, ancestors, depth-matched unrelated sites and all unrelated sites at unit dose. **C**, Cell-level localization for the first-order-current-matched additive perturbation, relation reassignment and the true focal shunt. Lines connect the same reconstructed cell. **D**, Localization across perturbation doses for shunting and the additive control; shading is the 95% cell-bootstrap interval. **E**, Focal-shunt localization across topological depth. **F**, Shunt-minus-additive localization across normalized synaptic conductance scales and inhibitory reversal potentials; the first three labels give the synaptic scale and the final two give E_I. **G**, Principal contrast for all mapped contacts and directly typed inputs. **H**, Exact factor-freeze control: post-shunt voltage and driving force are identical in both states, while only the full-shunt state includes the post-shunt adjoint transport field. **I**, Shunt-minus-additive localization after computing axial and leak conductances from cable geometry across specific membrane resistances. The x axis is the cell-median axial-to-leak ratio; circles and squares distinguish the original and v661 cohorts. Reassigned relations in **C** were stratified by topological-depth quartile when another site was available. Points and intervals in **C,F–I** are cell means and 95% cell-bootstrap intervals. The reconstructed cell is the unit of analysis; 101 focal sites are averaged within eight cells from one animal. At fixed relative dose, changing R_m changes both the electrotonic regime and the absolute shunt conductance; fixed-absolute, input-conductance-normalized and distributed-background tests are reported in Supplementary Fig. S11.

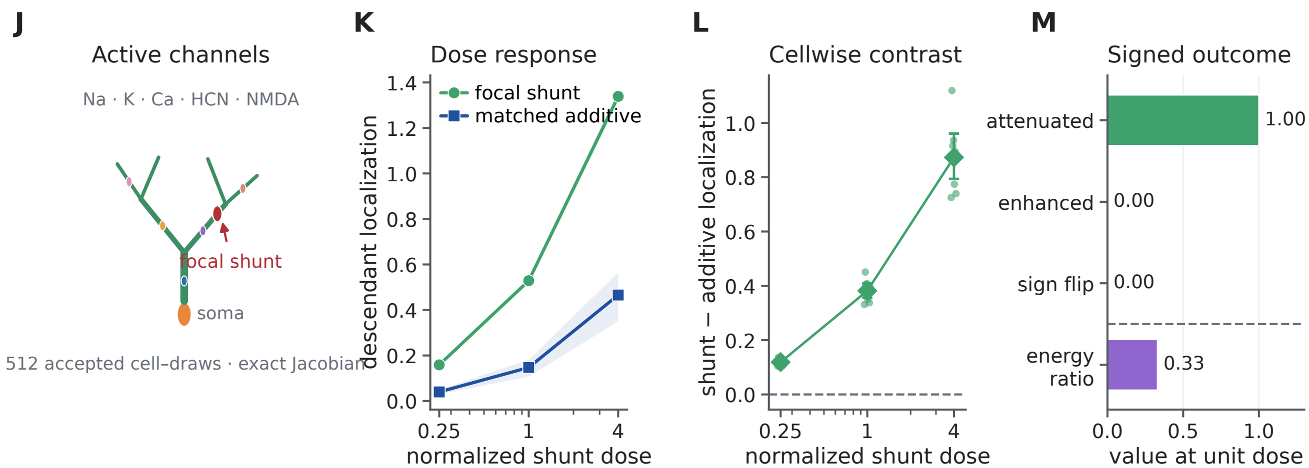

Figure 7: **The focal localization signature survives an active-conductance steady-state extension. J**, Na, K, Ca, HCN and NMDA conductances were sampled 64 times within each of eight reconstructed cells; every nonlinear equilibrium was solved before evaluating its exact local Jacobian. All draws used $R_m = 1{,}000\,\Omega\,\mathrm{cm}^2$ with one baseline-leak unit of distributed background conductance. **K**, Descendant localization across input-conductance-normalized doses for focal shunting and state-matched additive input. **L**, Paired shunt-minus-additive localization; small points are reconstructed-cell means after averaging channel draws and focal sites. **M**, Descendant-gradient outcomes at unit dose: attenuation fraction, enhancement fraction, sign-flip fraction and post-shunt-to-baseline energy ratio. Lines and bands in **K** and intervals in **L** are means and 95% cell-bootstrap intervals. Channel draws and sites are nested sensitivities; reconstructed cells are the units of inference.

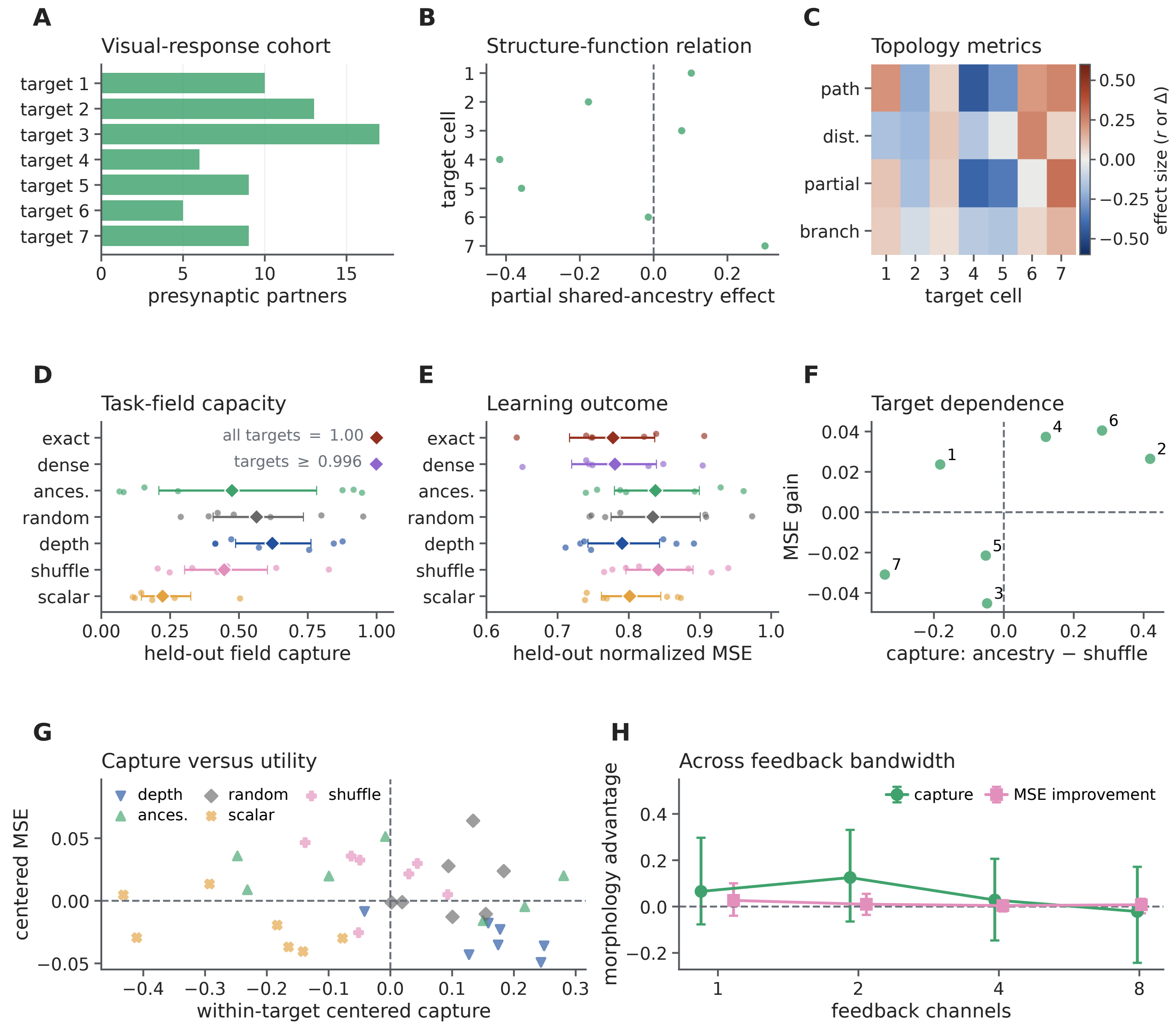

Figure 8: **A coarse major-branch visual-response surrogate does not reveal morphology-specific credit alignment.** **A**, Number of anatomically connected, functionally imaged presynaptic partners for each of seven target cells. **B**, Target-level partial Spearman ρ between functional similarity and shared ancestry after controlling for Euclidean separation and depth. **C**, Target-by-metric map of four structure–function statistics; blue denotes negative and red positive association. **D**, Held-out capture of task-derived branch credit with four feedback channels. **E**, Normalized held-out prediction error after learning from each projected field. In **D,E**, coloured points are target-cell means over ten fixed stimulus splits; diamonds and intervals show the mean and target-bootstrap 95% interval across seven cells. **F**, Per-target morphology-minus-shuffle capture and prediction-error improvements; labels identify target cells. **G**, Capture and learning outcome after centering both variables within target; the permutation test shuffles method labels within target. **H**, Ancestry-minus-shuffle contrasts across channel budgets. Capture is morphology minus shuffle, whereas mean-squared error (MSE) improvement is shuffle minus morphology; positive values favor morphology. Projection coefficients are fitted oracles and therefore upper bounds on each feedback family. Partners assigned to the same first major branch share one task-dependent error in this model; the analysis therefore cannot test fine within-branch credit.

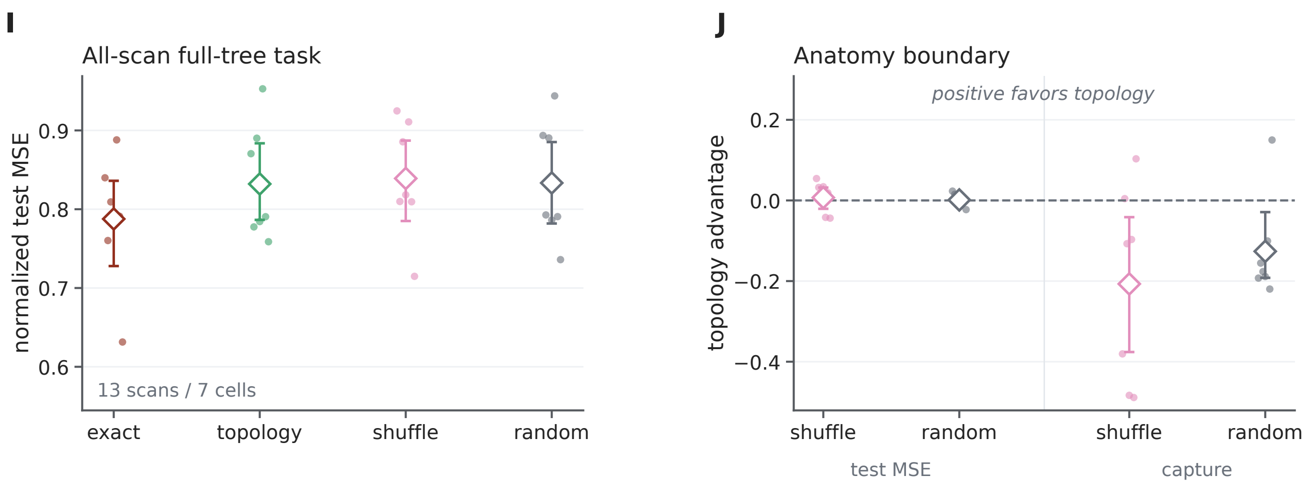

Figure 8: **A complete-tree measured-response task retains the anatomy– alignment boundary. I**, Normalized held-out MSE for exact compartment error and three equal-rank four-channel rules on complete reconstructed trees. Thirteen eligible scans and ten fixed splits per scan were averaged within seven target cells before inference. **J**, Per-cell topology advantage in learning (control MSE minus topology MSE) and common-checkpoint update capture (topology minus control). Positive values favor topology. Scan observations and split replicates are nested within target cell.

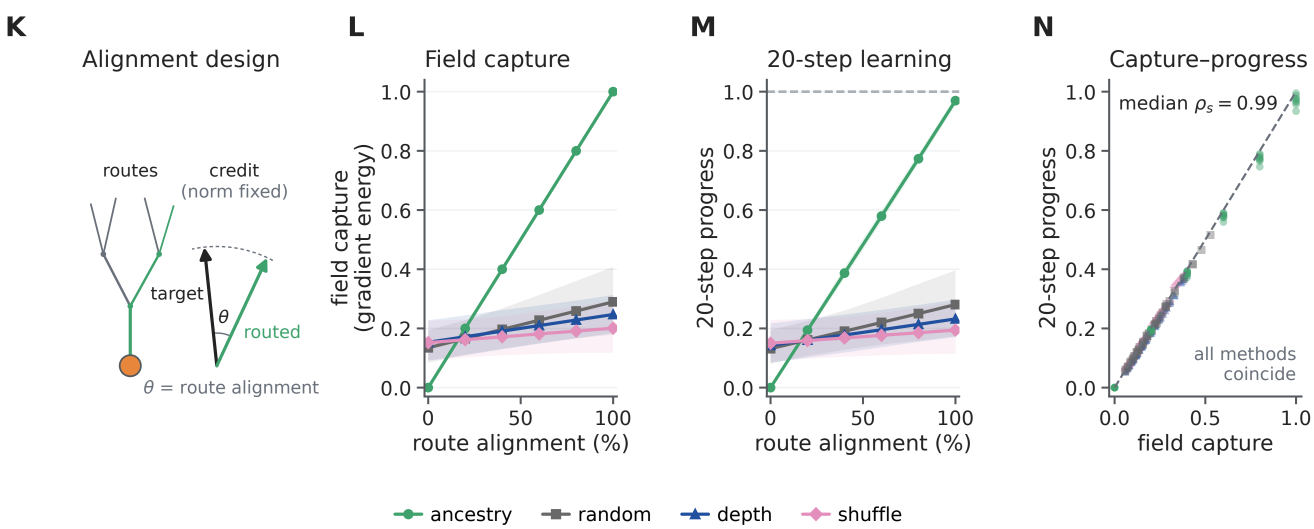

Figure 8: **Imposed task alignment converts the same fixed anatomy from an unhelpful to a useful feedback basis. K**, Controlled-alignment design: unit-energy exact gradients are rotated into or out of the fixed morphology dictionary at fixed curvature. **L**, Gradient-energy capture across imposed alignment for morphology-selected routes and three eight-channel controls. **M**, Relative 20-step loss reduction for the same fields and dictionaries. **N**, Initial capture versus 20-step progress; the annotation is the median within-cell Spearman correlation across the three control dictionaries. Lines summarize eight reconstructed cells, Monte Carlo streams are nested within cell, and bands in **L,M** are 95% cell-bootstrap intervals. This is a quadratic sufficiency test with oracle projection coefficients, not evidence that the cells used these gradients in vivo.

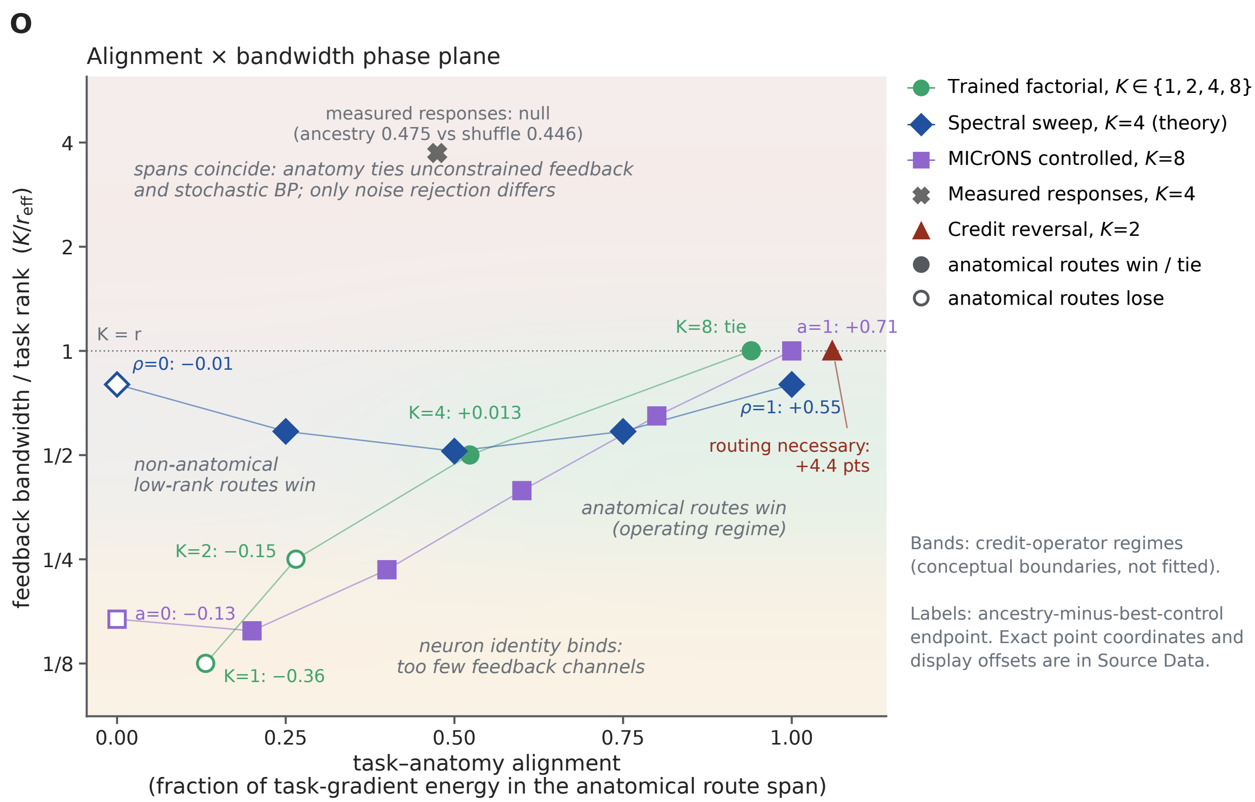

Figure 8: **Alignment and bandwidth organize the experimental boundary.** **O**, Task–anatomy alignment is the fraction of task-gradient energy inside the anatomical route span; normalized bandwidth is K/r_{eff}, where $r_{\text{eff}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$. Markers place the trained subtree factorial, the spectral phase sweep, the controlled MICrONS alignment test, the measured-response null and the two-stream credit-reversal task on the same plane. Labels report ancestry-minus-best-control endpoints; filled markers indicate a win or tie and open markers a loss. Shaded regions summarize credit-operator regimes and are conceptual boundaries rather than fits to the displayed outcomes. Exact coordinates and display offsets are in Source Data.

Discussion

Because each exact synaptic gradient separates into locally available eligibility—presynaptic activity, driving force and input resistance—and a transported compartment error, biological learning in a dendrite becomes a well-posed approximation problem: recover a defined credit field from restricted signals [21]. The present study extends that regular-tree result into a spatial hierarchy. Point-neuron theories ask how an error reaches a neuron or gates eligibility; a dendritic theory must additionally ask which subtree receives the coordinate and how strongly it is transported. Learning can therefore fail by losing neuron identity, addressing the wrong tree or descendants, or omitting route gain.

The prospective experiments separate these failures quantitatively. One coordinate per neuron closes most of the scalar-feedback gap, while deranging the neuron-to-tree map at matched bandwidth produces a smaller but reproducible cost. A 2,700-fit credit-reversal factorial then shows that the value of within-neuron structure is bandwidth dependent. Correct ancestry is inferior to unrestricted low-rank routes at one and two coordinates, acquires a small but reliable advantage over the best matched non-anatomical control at four coordinates, and ties at full rank. It also outperforms a degree- and depth-matched rewired tree at the intermediate budgets. Dendritic, flat, grouped-point and explicitly gated point implementations are identical when given the same routed fields, locating the benefit in structured address resources rather than dendritic material per se. Exact transported feedback makes the derived local eligibility sufficient to approach matched backpropagation, and the detached clean audit found no material exact-transport–backpropagation difference, bounding the claim to empirical agreement rather than formal equivalence. Gradient alignment predicts immediate descent.

The credit-operator bound now unifies these observations. Route quality is not capture alone: it is retained task signal divided by finite-step gain and admitted stochastic noise. This utility predicted the existing factorial better than spectral capture alone. In the independent phase experiment, ancestry became useful as task covariance rotated into its span; routed depth was best at the task's hierarchy depth; and a projection improved a noisy gradient only below the rejected-noise/discarded-signal boundary. These are not routes by which a local rule can beat exact full-batch backpropagation on the same objective. They explain how a restricted stochastic rule can improve relative to another noisy or bandwidth-limited rule, with backpropagation plus the same projection remaining the relevant upper control. The interior-optimum reanalysis sharpens this division of labour: the bound's optimum located the trained depth optimum wherever one existed, whereas the trained $K = 4$ anatomical advantage lies beyond one-step geometry (Supplementary Fig. S16). The results therefore order into four comparisons. A restricted rule never beats exact full-batch backpropagation at matched step norm; it beats stochastic backpropagation exactly when rejected noise exceeds discarded signal; it beats a bandwidth-matched non-anatomical rule only when task covariance aligns with the anatomical span; and any within-tree address beats scalar or neuron-only feedback whenever exact credit varies across the tree in an addressable way. Reliability-weighted route gains improve each comparison when branch signal-to-noise is heterogeneous. Figure 8o places every bandwidth-varied experiment of this study in the resulting alignment-by-bandwidth plane, including the measured-response null, which falls where the theory predicts a null. In operator terms, topology sets the span of addressable fields, conductance sets the metric that weights their use, and the task determines whether either matters.

Depth can affect learning in at least three distinct ways. First, nested branch points reuse route coordinates across exponentially many leaves and provide progressively finer addresses; the matched-depth phase result isolates this role. Second, serial dendritic stages can compose nonlinear forward operations, a role deliberately absent from the quadratic phase experiment. The exact-resource positive-rate experiment now isolates one such case: aligned divisive composition improved accuracy by roughly 31 percentage points in both $H = 2$ and $H = 3$ variants, whereas literal grouped-point models stayed flat and reversed placement made serial depth harmful. Independent and shuffled-sensor controls removed the H3 gain, and the raw-additive series worsened with depth. This is evidence for matched hierarchical shunting within the calibrated task family, not a general law that deeper trees are better. Third, a deeper physical route transports a shared coordinate through more differentiated branch states. Consistent with that cost, shared-soma gradient cosine fell from 0.946 at D1 to 0.767 at D3, whereas exact path transport reproduced autograd at every checkpoint. The earlier additive fixed-contact control likewise showed no intrinsic depth benefit when terminal branches and contacts were fixed. Thus the dominant requirement remains a signed coordinate for the correct neuron, followed conditionally by enough task-aligned address and computational depth—not depth in isolation. The new controls make the point–dendrite boundary explicit: the serial tree beat a resource-identical nonserial star only under the aligned hierarchy; it also beat the stricter literal grouped-point model at both H2 and H3, yet ordinary parameter-matched point MLPs beat the tree by about seven points. Dendritic depth is therefore a structured inductive and wiring resource in this task, not an absolute expressive separation. The credit ladder draws the complementary BP–local boundary: soma-broadcast autograd nearly matched BP under a common optimizer, optimizer choice accounted for the largest raw separation from shared LocalCA, and path specificity recovered most of the remaining local-rule deficit. Stated as an accounting at matched resources, the tree's contribution separates into a forward-composition term isolated by the backpropagation depth contrast, an address-resolution term isolated by the bandwidth factorial, and a transport cost isolated by the falling shared-coordinate cosine, with the local-rule retention measuring their interaction.

Real dendrites then constrain how that coordinate could be distributed. Reconstructed arbors compress model-matched and independently generated cable fields into sparse, ancestry-defined supports using a small fraction of dense feedback wiring—in wiring-normalized terms, an order-of-magnitude advantage in captured energy per feedback connection over a dense oracle, of which a 2.7-fold factor is anatomy-specific at matched density (Fig. 6k,l). A dendritic tree, on this account, multiplexes K addressable fields within one neuron's arbor. An equivalent point implementation requires the same K separately addressable teaching coordinates; its wiring cost depends on how those coordinates are routed. Yet fine morphology does not reliably outperform controls matched for depth and parent degree. The synapse-resolved inhibitory census similarly supports coarse co-targeting and class-specific spatial scales, while shared-path effects do not survive axon-level aggregation and joint geometric matching. These one-mouse measurements establish available addresses, not their use for endogenous learning.

Shunting has a precise and bounded role within this hierarchy. Unlike models in which inhibitory circuits compute an error [38–40], focal conductance here changes the sensitivity of routes crossing a compartment. Exact factor freezing assigns the positive localization effect in the normalized proxy to altered adjoint transport, partly opposed by the change in synaptic driving force. Physical calibration shows that the effect can become very small in passive regimes with high axial-to-leak coupling and re-emerge in higher-conductance states. This spatial boundary is determined by the focal Green's-function column and cable length constant: strong axial coupling spreads the perturbation and reduces the descendant-versus-matched off-path contrast. The passive inverse-operator corollary is used only as a global magnitude bound, $\|\Delta G^{-1}\|_2 \leq \kappa \|G^{-1}\|_2^2/[1+\kappa/\lambda_{\max}(G)]$; it does not predict spatial selectivity. Shunting is therefore a state-dependent route-gain mechanism, not a universal optimizer and not evidence that inhibition itself carries a teaching signal. The active-channel ensemble extends this conclusion beyond a passive operator: exact linearization at 512 accepted Na/K/Ca/HCN/NMDA steady states preserved descendant-enriched attenuation without gradient sign reversal. It remains a local steady-state result, not a claim about channel kinetics or spiking. The reliability phase identifies a possible computational use for that attenuation: fixed-step branch gains $\min\{1, S_b/[c(S_b+N_b)]\}$ suppress unreliable credit and outperform one global gain when reliability differs across branches. The state-matched positive-conductance experiment joins that prediction to a physical input-resistance mechanism in a trained branch model: aligned shunts improved the immediate credit step and final loss over a global shunt, but not final loss over unshunted noisy learning. The independently computed point-gate control agrees to numerical precision and proves that the operation is not mathematically exclusive to dendrites. This differs from the sibling gain–load study, which asks when branch-local inhibition improves forward signal robustness; here the forward state is matched and conductance weights backward credit [41]. Moreover, branch reliabilities and shunt placement were supplied by an oracle, and the compensating current removed forward state changes. The experiment therefore validates a conductance-based, state-dependent approximation to a reliability factor, not autonomous inference of that factor or a completed biological causal chain.

Anatomical capacity is useful only when it matches the task. Neither all 13 eligible visual-response scans nor 520 learning fits on the complete compressed segment trees reveal a reliable morphology-specific advantage for the derived objective. By contrast, rotating exact task credit into or out of the same anatomical span reverses its learning benefit. The six-animal reanalysis supports the availability of signed neuron-specific coordinates, but does not identify their transport within a tree. Route overlap and off-route leakage then predict one-step interference (Supplementary Fig. S6). Together, these results replace the claim that more dendritic structure is inherently better with a conditional statement: route capacity matters when the task requires credit fields that the arbor can express.

The alignment requirement has a population-level analogue: motor learning is faster for perturbations within an existing neural manifold than outside it [42]. Our claim is the subcellular counterpart, not an identification of the population mechanism: a pre-existing ancestry span helps only when the task-gradient field lies sufficiently within it.

These results relate to, and delimit, three strands of prior work. Studies of dendritic nonlinearity establish that branch-local processing expands a single neuron's forward capacity [43–46]; here forward resources were deliberately matched or budgeted, and the tree's contribution was measured instead in feedback bandwidth, so the two accounts are complementary rather than competing. Dendritic-error and segregated-compartment learning models deliver a teaching signal to one or a few named compartments [17–19, 47]; the present framework asks how far such a signal can usefully be distributed within a full arbor, and answers with an explicit bandwidth–alignment condition rather than an architectural preference. Observations of branch-specific plasticity and structured synaptic placement [48–50] demonstrate that learning is spatially organized in vivo; our analysis specifies what that organization must deliver to constitute credit routing rather than forward gain control—nested addresses whose span aligns with task-gradient covariance—and provides the perturbation signature that distinguishes the two.

Bio-inspired machine-learning systems likewise use explicit context gates, active dendritic segments or localized dendritic updates to allocate learning capacity [51–53]. Those algorithms establish useful engineered routing schemes; the present theory instead predicts when a restricted biological route should help or fail at a fixed bandwidth.

The same accounting suggests why credit might be routed through trees at all. With a fixed budget of routed teaching coefficients, a circuit can spend them as many neurons carrying one coordinate each or as fewer neurons with finer within-tree addresses; the implementations are mathematically interchangeable, but their wiring is not. Per-neuron coordinates require long-range feedback axons, whereas per-branch routes reuse local inhibitory synapses on existing arbors, so the wiring-efficient allocation depends on whether task credit varies mostly across neurons or hierarchically within them. On this view the tree is the inexpensive way to buy feedback bandwidth—a question of circuit economics rather than computational exclusivity.

The framework makes a branch-resolved experimental prediction. After matching somatic voltage, output and global error, focal dendrite-targeting inhibition during learning should alter subsequent plasticity more strongly for spines below the perturbed branch than for equidistant spines on sister subtrees. The contrast should scale with local input resistance and perturbation conductance, and a current-matched additive perturbation should not reproduce it. Simultaneous dendritic recording, focal inhibition and longitudinal spine measurements could distinguish credit routing from forward gain control [37, 48].

The theory assumes steady-state, non-spiking trees with unique directed paths; the reconstructed-cell analyses are sensitivity models rather than cell-specific physiological fits. The primary, sensitivity and inhibitory cohorts come from one mouse, the functional cohort contains seven targets, projection and transport coefficients are oracles, and the external animal analysis is retrospective. The trained within-neuron factorial uses one synthetic eight-context hierarchy, analytically controlled route fields and implementation-equivalent linear parameterizations. The active extension linearizes steady states and does not simulate channel kinetics. Temporal eligibility, spikes, recurrent paths and long-term network adaptation remain outside the current model. In particular, branch plateau potentials and behavioral-timescale synaptic plasticity provide documented dendritic teaching signals beyond this steady-state theory [54, 55]. The hierarchy-depth phase uses a fixed-state quadratic family and prescribed task covariance. The nonlinear physical-depth bridge now tests $H = 2$ and $H = 3$, two fixed sensor inventories and one calibrated operating-point family; it does not cross the full $D_{\mathrm{r}} \times H \times K \times \rho$ design, infer sensor alignment or establish prevalence beyond this synthetic task. The positive-conductance reliability experiment trains excitatory branch weights but fixes oracle shunts and uses a compensating current clamp; it does not demonstrate autonomous estimation or learning of shunt placement. We establish no memory, latency or benchmark advantage over backpropagation. Within these limits, the result is an information hierarchy for cellular learning: coordinates identify neurons, ancestry partitions synapses, conductance regulates routes and task alignment determines utility.

Methods

Directed conductance-tree network

Each compartment combines nonnegative excitatory and inhibitory synaptic conductances, leak, and child-to-parent dendritic coupling. Excitatory and inhibitory reversal potentials were fixed within an experiment. Raw trainable parameters were transformed to nonnegative conductances. A compartment could apply a monotone branch output nonlinearity $a_n = f_n(V_n)$ before transmitting activity to its parent; this mechanism is named "reactivation" in the implementation, and f_n was the identity when it was disabled. The full forward equations, raw-parameter derivatives, and temporal order of the local update are provided in Supplementary Section S1.

Writing $p(n)$ for the parent of compartment n, the compartment-error recursion follows from the chain rule on the tree:

$$\frac{\partial \mathscr{L}}{\partial V_n} = \frac{\partial \mathscr{L}}{\partial V_{p(n)}} f_n'(V_n) R^{\text{tot}}_{p(n)} g^{\text{den}}_{n \to p(n)}. \tag{15}$$

Multiplying this recursion along the unique path yields Eq. 7. The local derivative of Eq. 5 with respect to g_i yields Eq. 6. Analytical gradients were checked against PyTorch automatic differentiation on fixed batches and checkpoints. Parameter tensors, rather than individual elements, were the units of the reconstruction diagnostic.

Local learning and feedback fields

The three-factor update was the batch average of

$$\widehat{\nabla}_{g_i} \mathscr{L} = x_i R^{\text{tot}}_n (E_i - V_n) \widehat{\delta}_n. \tag{16}$$

The tested $\widehat{\delta}_n$ fields included a global scalar, the matched-width/scalar-fallback implementation, neuron-indexed sharing, low-rank random feedback, selected paths, and exact path transport. The neuron-indexed field repeats one somatic coordinate over all descendants of that soma. It does not reveal an independent error at each compartment. When $\widehat{\delta}_n = \partial \mathscr{L} / \partial V_n$, the three-factor rule equals the exact conductance gradient; the exact-transport experiment therefore isolates feedback approximation from local eligibility. Four- and five-factor variants multiply the three-factor update by bounded, slowly estimated stage-level scalar reliability terms. They were treated as preconditioners rather than theoretically required teaching signals.

Regular-tree feedback comparisons used MNIST classification with matched $[3,3]$ morphologies, optimizers, decoders, schedules, and paired initializations within architecture. The feedback-definition experiment used 15 independently initialized seeds per architecture and condition. Exact transport used five matched shunting seeds per rule and decoder-update condition. Test data remained hidden during checkpoint selection. Full hyperparameters and provenance are reported in Supplementary Section S5 and the accompanying machine-readable configurations. This feedback-definition cohort is a re-execution under the released implementation with the width- and connectivity-matched additive architecture; the original conference cohort gave $90.75 \pm 1.01 \to 97.15 \pm 0.12$ (shunting) and $91.58 \pm 0.61 \to 97.14 \pm 0.10$ (additive), consistent with the values reported here [21].

The Fashion-MNIST replication changed only the dataset and used the same regular $[3,3]$ architecture, 128 somata, contact counts, three-factor rule, local decoder, architecture-specific initialization, optimizer and validation-based checkpoint selection. Images were flattened to 784 inputs without normalization or augmentation. Within each fresh seed 10200–10209, the 60,000-image training set was split into 48,000 training and 12,000 validation examples; the official 10,000-image test set was used only after selection. Sixty fits crossed two architectures with scalar fallback, neuron-indexed feedback and exact path transport. The primary replication criterion required the paired neuron-indexed-minus-scalar interval to exclude zero and at least 8/10 positive seed signs in each architecture. Intervals used 50,000 paired-seed bootstrap draws and exact two-sided sign-flip tests enumerated all 2^{10} assignments. The artifact gate required complete finite results and checkpoints, the registered seeds and no fallback warning; all gates passed. A frozen-source guard divided each architecture across two source manifests after a concurrent edit. The exact diff only added an optional rule override that no resolved configuration enabled, and the audit retains both manifests.

Regular-tree regime and one-step analyses

The task, inhibition-dose, depth, broadcast-noise, rule-family, error-source, mechanism-control, feedback-ladder, and CIFAR-10 analyses use the frozen source tables and configurations from the original regular-tree study. Unless stated otherwise, each condition contains five independent training seeds. The three-factor, four-factor, five-factor comparison and the local-mismatch control contain three seeds and are treated as descriptive. The context-gating and noise-resilience tasks are synthetic classification tests described in Supplementary Section S5. CIFAR-10 images were flattened; no convolutional front-end, augmentation pipeline, or large-scale benchmark claim was used.

The prospective depth-by-feedback cohort was frozen before outcome inspection. It crossed two tasks, two neuron models, four dendritic depths, and three local feedback fields at ten seeds, together with one matched backpropagation model per task–core–depth–seed combination (480 local-learning and 160 backpropagation runs). The local conditions used the factorization-derived three-factor rule. Configurations fixed the decoder, optimizer, training horizon, reactivation initialization policy, 40 excitatory and 20 inhibitory contacts per distal branch, and one population of 128 modeled neurons. The single-population design prevented equal neuron indices in different feedforward populations from sharing a teaching coordinate. Nominal depth changed compartment and parameter count; it was therefore treated as a depth stress test rather than a capacity-matched comparison.

Each frozen run had to contain its resolved configuration, seed record, checkpoint, complete learning curves, resource record, and finite held-out metrics. The artifact audit also required every expected dendritic stage and excluded critical log errors, non-finite parameters, degenerate coupling stages, and transport-shape fallbacks. A subsequent outcome-independent input- validity rule excluded the 160 synthetic-noise shunting runs because a signed transfer output cannot parameterize positive conductances. This left 480 publication-facing runs. Condition means used the ten paired seeds. Intervals for paired contrasts used 20,000 bootstrap resamples, and exact two-sided Wilcoxon tests were computed from the ten seed differences.

The bandwidth-matched routing cohort was frozen separately and crossed the two tasks and neuron models with depths two and four, correct neuron-indexed routing or a fixed derangement, and seeds 42–51. The derangement maps every somatic coordinate to another neuron and has no fixed point. It preserves the number and per-example distribution of teaching coordinates, the forward network, and all training settings. The same validity rule removed 40 synthetic-noise shunting runs, leaving 120 runs and six task–model–depth conditions. Correct-minus-deranged accuracy was paired by seed. We report paired bootstrap intervals, exact two-sided Wilcoxon tests, and Benjamini–Hochberg correction across the six retained contrasts.

Trained within-neuron subtree-address experiment

The two-stream support-task configuration, task parameters, ten confirmatory seeds and artifact gates were frozen before the canary outcome was inspected. Each example contained two 12-dimensional sibling-stream inputs, a binary context and a balanced binary label. For the context-selected stream, the input mean was the label sign times a seed-specific unit teacher vector; the other stream had the opposite sign at 0.75 times the amplitude. Independent Gaussian noise with standard deviation 1.15 was then added. Every train and test split contained identical counts for all four context–label combinations. Thus context had zero label correlation, both streams were active, and the inactive stream was a structured distractor rather than a zero mask.

For branch weights $\boldsymbol{w}_A, \boldsymbol{w}_B$, the somatic logit was

$$z = \mathbb{1}[c=0]\boldsymbol{w}_A^{\top}\boldsymbol{x}_A + \mathbb{1}[c=1]\boldsymbol{w}_B^{\top}\boldsymbol{x}_B. \tag{17}$$

All conditions used the same paired initialization, mini-batches and forward model. The one-neuron and depth-shared fields applied the logistic output error to both branches. Correct $K=2$ feedback applied it only to the selected sibling, while fixed derangement applied it to the other sibling. The random rank-two control rotated the two context coordinates by a seed-fixed orthogonal matrix. Exact transport and analytic backpropagation used the selected-branch derivative. The gated point emulation used the same two parameter groups and context-gated input eligibility; it therefore represents the explicit point-model resource that reproduces the dendritic address.

Each seed used 2,048 training and 2,048 independent test examples, mini-batch SGD with batch size 64 and learning rate 0.3 for 60 epochs, and Gaussian weight initialization with standard deviation 0.03. Switch interference was measured after a further 12 epochs on 1,024 balanced-label examples from context 1 only, without resetting optimizer or weights. Gradient cosine and scaled capture used the concatenated per-example parameter gradients. For one-step progress, each candidate batch gradient was norm matched to the exact gradient before applying a fixed step of 0.25. Canary seed 4099 was used only to enforce balance, nondegenerate capture, minimum exact-route accuracy and exact–BP equivalence. The unchanged code and configuration then ran confirmatory seeds 4100–4109. The independent seed was the paired inferential unit; intervals used 20,000 paired bootstrap resamples and exact two-sided Wilcoxon tests are reported in Source Data. This experiment tests $K = 2$ on one shallow matched forward architecture and served as the support check for the larger factorial.

The full subtree-address protocol used eight contexts assigned to the leaves of a balanced binary hierarchy and eight eight-dimensional input streams. Every context–label cell was exactly balanced. The selected stream carried unit label signal; non-selected streams carried an opposite-sign distractor whose amplitude increased with tree distance (0.15 for a sibling, 0.45 within the same half and 0.75 across halves), followed by independent Gaussian noise of standard deviation 1.1. Each seed supplied 2,048 train, 2,048 test and 1,024 switch examples. Full-batch gradient descent used learning rate 0.3 for 80 epochs and 20 switch epochs. Confirmatory seeds 4200–4219 were fixed before outcome inspection; canary seed 4199 was excluded from inference.

Five representations were crossed with $K = 1, 2, 4, 8$: the task-matched dendritic tree; an explicitly context-gated point model; a flat compartment model; grouped point subunits; and a degree- and depth-matched tree with a seed-fixed leaf permutation. The first four implementations deliberately received identical routed coefficient fields and are computationally equivalent controls. Every representation contained 64 trainable forward weights, 64 active input contacts, one output nonlinearity and one scalar decoder. Route fields were normalized to unit Euclidean norm for each context. The six families were correct contiguous ancestry subtrees, a fixed within-neuron route derangement, depth-interleaved bins, random sparse groups, an unrestricted random rank-K projector and a training-derived rank-K PCA upper bound. Shared neuronal feedback, exact compartment transport and analytic backpropagation supplied references. Random controls were keyed only by seed, family and bandwidth, ensuring identical stochastic realizations across implementation-equivalent representations.

The frozen canary required exact context–label balance, zero context–label correlation, neuron-shared capture below 0.20, exact capture above 0.999999999, exact–backprop endpoint difference below 10^{-12}, exact implementation equivalence below 10^{-12} and zero fallbacks. All gates passed before the 2,700 confirmatory fits ran with unchanged script and configuration hashes. Outcomes were paired by seed. Intervals used 20,000 paired bootstrap draws; two-sided Wilcoxon tests used the 20 seed differences. The per-seed best matched non-anatomical control was prespecified across route derangement, depth-interleaved, random-sparse and random-rank conditions; the learned PCA route remained an explicitly labeled upper bound.

Credit-operator phase experiment and frozen-factorial reanalysis

The phase experiment used 16 Euclidean coordinates and the orthonormal Haar basis of a balanced binary tree, ordered from the global mode (level 0) to single sibling contrasts (level 4). Its configuration, 50 confirmatory seeds 7000–7049, two artifact-only canary seeds and numerical tolerances were frozen before confirmatory outcomes. The canary required basis orthogonality, projector idempotence, static-gain span invariance and exact agreement of the explicit point gate. Confirmatory execution required unchanged SHA-256 hashes of the configuration, runner and numerical reference. Every expected row was present: 3,750 spectral, 1,800 depth, 3,750 projection and 1,500 reliability rows.

For the spectral screen, task covariance assigned total energy $(0.15, 0.35, 0.30, 0.20, 0)$ to tree levels 0–4. An alignment coefficient $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$ mixed this covariance with an independently rotated covariance having the same eigenvalues. Ancestry, random rank-matched and dense principal-eigenspace projectors were compared at $K \in \{1, 2, 4, 8, 16\}$. The dense value was also evaluated as the Ky–Fan sum of the leading K eigenvalues divided by total covariance energy. Because the two endpoint covariances had equal trace, ancestry-minus-random capture was affine in alignment; endpoint differences supplied the exact per-seed crossover, with already favorable zero-alignment seeds assigned threshold zero. Capture was $\mathrm{tr}(P\Sigma_g)/\mathrm{tr}(\Sigma_g)$. The same-span diagnostic compared raw nested indicators, their tree-Haar basis and indicators multiplied by fixed nonzero lognormal column gains; rank, coherence, positive-spectrum Gram condition number and capture were computed directly.

For the depth screen, a unit-norm target was sampled from modes at levels 0 through H, with $H \in \{1,2,3,4\}$. Gradient-noise standard deviation in level ℓ was $0.65[1+1.5\max(\ell-H,0)]$. Starting from zero, 80 common-noise updates of size 0.12 used the projector onto levels 0 through $D_{\mathrm{r}} \in \{1,2,3,4\}$; a fixed leaf permutation supplied the rewired control and the identity supplied stochastic backpropagation. The projection screen independently set retained signal r and retained noise n to $\{0.25, 0.5, 0.75, 0.9, 1\}$ and compared an unprojected stochastic gradient, backpropagation plus the same projection and a routed estimator with an added 0.05 total coefficient-noise fraction. At the unit step and identity-curvature quadratic, their expected losses were calculated analytically and stochastic losses were recorded from paired draws.

For the reliability screen, eight disjoint blocks had signal and noise energies proportional to $\exp(hx_b)$ and $\exp(-hx_b)$, respectively, with equally spaced $x_b \in [-1,1]$ and heterogeneity $h \in \{0, 0.5, 1, 1.5, 2\}$. We compared no attenuation, the best common gain, $S_b/(S_b+N_b)$, seed-shuffled and reversed gains, and a point gate receiving the identical branch factors. Intervals resampled the 50 paired seeds with 20,000 draws; paired tests were two-sided Wilcoxon signed-rank tests.

The secondary analysis reconstructed the frozen training data, initialization and route matrices for all 2,700 previously completed subtree-factorial fits. For each of 20 seeds, 64 independently sampled minibatches of size 64 estimated gradient-noise energy at initialization. The exact full-data gradient supplied signal, the largest logistic Hessian eigenvalue supplied local smoothness, and Eq. 9 supplied utility. Correlations were restricted to the 540 dendritic-tree route conditions and uncertainty resampled whole seed blocks. This is an initialization diagnostic on an existing experiment, not a second confirmatory training cohort.

Positive-rate nonlinear physical-depth experiment

To connect the fixed-state hierarchy result to the original rate-based network, we used the registered `hierarchical_gain_load` generator and the production population-network DendriNet. Inputs were ordered as nonnegative excitatory and inhibitory rate streams. For binary label y, the distal excitatory signal had the schematic form

$$E_j = (e_0 + y\Delta + \varepsilon_j)\exp\left\{\sum_{\ell=1}^{3} \sigma z_{\ell,a_\ell(j)} - \frac{1}{2}\sum_{\ell=1}^{3}\sigma^2\right\}, \tag{18}$$

where $a_\ell(j)$ indexes the fine, coarse or global latent factor containing coordinate j. Three disjoint inhibitory blocks separately measured those factors. With sensor alignment α, their log gains were $\sigma[\alpha z_{\ell,a_\ell(j)} + \sqrt{1-\alpha^2}z'_{\ell,a_\ell(j)}]$. Thus $\alpha = 1$ supplied matched branch-local nuisance sensors, whereas $\alpha = 0$ preserved the complete factor marginals with independent sensors. A trial-shuffled control cyclically permuted the sensor field after class balancing, preserving its marginal distribution while removing example-wise pairing.

At every physical stage the production shunting equation was

$$V_n = \frac{E_n + C_n}{1 + E_n + I_n + G_n + \varepsilon}, \tag{19}$$

where C_n and G_n are child current and child conductance. All rate and conductance terms were nonnegative. Reactivation was the identity, so the divisive branch equation was the only intermediate nonlinearity. D1 [8], D2 $[2,3]$ and D3 $[2,1,2]$ had one, two and three nonsomatic stages, respectively, but each retained eight nonsomatic branch units per soma and the same 4/2/2 fine/coarse/proximal sensor inventory. Every model contained 64 somata, 16 excitatory and 12 inhibitory contacts per branch, 14,336 active synapses, 21,760 candidate slots, 66,178 trainable parameters and 2,944 persistent state scalars per sample. Reversing the fine and global feature ranges provided the same-resource tree-placement control.

The operating point used signal contrast $\Delta = 0.80$, matched per-factor train/test log-gain standard deviation 0.25, child conductance 16 and mechanism-neutral initialization. It was selected transparently after an original severe-shift canary failed at every depth and two-seed exploratory gain, coupling and signal ladders located an accessible non-ceiling boundary (Supplementary Fig. S15). No pilot seed entered inference. The fresh confirmatory seeds 10200–10209 crossed aligned, zero-alignment, trial-shuffled and reversed-placement shunting BP; aligned raw additive BP; and aligned or reversed-placement shunting LocalCA with shared-soma or exact path transport, for 270 fits. Adam ran for at most 180 epochs with validation-based early stopping. Learning rate was 0.003 for BP and 0.0008 for LocalCA; all comparisons were paired by seed.

The prespecified primary effect was aligned D3 minus D1 accuracy. Topology specificity required the aligned depth effect minus the corresponding zero-alignment, shuffled-sensor or rewired-tree depth effect. A positive local claim separately required its depth effect and aligned-minus-rewired interaction. Intervals used 50,000 seed-paired bootstrap draws; exact two-sided sign-flip tests enumerated all 2^{10} paired sign assignments. The artifact gate required all 270 finite results, no LocalCA fallback message, exact resource equality, at least nine of ten aligned BP seeds above 0.55 at every depth and condition means below 0.98. A dedicated checkpoint replay measured branch voltage, total conductance, input resistance, activation derivative, path-gain dispersion and local-versus-autograd conductance-gradient cosine on a fixed held-out batch.

Point–dendrite and BP–local-credit controls

The same frozen generator, operating point, stopping rule and paired seeds were used for a 140-fit reviewer-requested extension. The all-active grouped-star model retained the hierarchical model's complete state dictionary and resource counts but evaluated every nonsomatic level independently and pooled the resulting child aggregators only once at the soma. D1 is therefore an equality control; D2 and D3 remove serial composition without removing branch modules. The aligned and reversed-placement regimes were crossed with D1–D3. Two ordinary point MLPs were trained on the aligned task: one matched the active parameter count (15,055), and the nearest integer-width model matched the total count (66,200 versus 66,178 in the tree). Neither MLP was supplied the task hierarchy or branch grouping.

For the teaching-coordinate control, the forward pass and synaptic autograd derivatives were unchanged, but a backward hook replaced every nonsomatic output derivative by the derivative at its owning soma. This "soma-broadcast autograd" condition was first run with the standard BP optimizer, which makes its contrast with full BP coordinate-specific. A configuration audit performed after those outcomes revealed that it could not be compared mechanistically with LocalCA, which used split parameter groups and learning rates selected for its stability. Before viewing the amendment outcomes, we therefore froze 60 additional aligned and reversed D1–D3 fits that changed only soma-broadcast to the LocalCA optimizer groups (global rate 0.0008, branch-weight rate 0.0007 and decoder rate 0.0015). This amendment is identified as a post-outcome diagnostic, not a prospective rescue of the primary analysis.

The architecture, alignment-interaction and credit contrasts were computed within seed. Intervals used 50,000 paired-seed bootstrap draws, and exact two-sided sign-flip tests enumerated all 2^{10} assignments. The artifact gate required 200 finite new fits, no fallback pathway, seeds 10200–10209 and exact equality of trainable parameters, active contacts, candidate slots and persistent state between serial and star models. The implementation tests also required identical D1 forward logits and exact equality of forward logits between full and soma-broadcast autograd modes before training. All gates passed. Configurations, the dated primary contract and its amendment, seed-level outcomes and the machine-readable audit are included with the source data.

Physical-depth alignment dose response

The interpolation extension retained the same generator, shunting-BP model, operating point, early-stopping rule, resources and paired seeds as the physical-depth endpoint experiment. It crossed $\alpha \in \{0.25, 0.50, 0.75\}$ with D1–D3 for 90 new fits and combined them with the unchanged $\alpha = 0$ and 1 endpoint outcomes. The dated contract was frozen after the endpoint outcomes were available but before any intermediate run was inspected. Its primary summary was the ordinary least-squares slope, within seed, of D3-minus-D1 accuracy on alignment; the $\alpha = 0.75$ minus 0.25 change and adjacent-level changes diagnosed linearity. Intervals used 50,000 paired-seed bootstrap draws, and exact two-sided sign-flip tests enumerated all 2^{10} assignments.

A concurrent source-tree edit interrupted 26 of the first 90 scheduled jobs through the frozen-source guard.

Those jobs were rerun under the new source hash; the 64 completed pre-edit jobs were retained because the only code difference introduced an optional rule override that no resolved configuration enabled. A machine-readable two-manifest audit and the dated source-equivalence record accompany the source data. The final gate required exactly 90 unique, finite intermediate outcomes, no fallback pathway, the expected seeds and exactly equal resources across depth; all conditions passed.

Literal grouped-point and second-hierarchy experiments

After the original $H = 3$ endpoints, nonserial-star controls and alignment-dose outcomes were known, we froze a final 220-fit matrix before opening any new outcome. The literal grouped-point model retained each nonsomatic branch module, structured input mask, local shunting equation and trainable scalar, but replaced the serial child aggregator at every level by a direct parameter-matched projection to its owning soma. Each replacement contained the same number and initialized value of raw conductance parameters as the aggregator it replaced. Every nonsomatic level therefore operated on the raw external streams without receiving another branch state. Unit tests required finite trainable gradients, copied conductances and exact D1 logits against the serial model.

The H3 arm crossed aligned and reversed placement, D1–D3, BP and the existing paired seeds 10200–10209 for 60 new fits; the already observed serial cohort was retained unchanged as its paired reference. The H2 arm used two 32-feature sensor tiers, fixed inventory counts 4/4 and exact-resource morphologies D1 [8] and D2 $[4,1]$. It crossed aligned and reversed placement with serial BP, literal grouped-point BP and serial LocalCA using shared-soma or exact-path transport over ten fresh paired seeds 10300–10309, for 160 fits. Task-family, signal, gain, coupling, learning-rate, early-stopping and all other selected operating-point settings were inherited without retuning from H3.

Primary estimands were the H3 aligned serial-minus-grouped-point contrast at D3 and its placement interaction; the H2 serial-BP D2-minus-D1 effect and its placement interaction; the H2 serial-minus-grouped-point contrast at D2 and its placement interaction; and the H2 shared- and path-LocalCA depth effects and placement interactions. Intervals used 50,000 paired-seed bootstrap draws, and exact two-sided sign-flip tests enumerated all 2^{10} assignments. A directional positive claim required an interval excluding zero and at least eight of ten positive differences. All 220 fits completed with finite metrics, no fallback or nonfinite alert, the expected seeds and exact resource equality. Frozen manifests, run-level outcomes, every paired contrast and the machine-readable audit are distributed with Source Data.

State-matched positive-conductance reliability experiment

The locked mechanism experiment used eight branches with four nonnegative rate inputs per branch, 512 training examples, 2,048 test examples and one learned positive excitatory conductance per input. Branch voltages were $V_b = g_b^E/(g_b^L + g_b^E)$. A nonnegative shunt κ_b with reversal zero was paired with the oracle current $\kappa_b V_b$, so the post-intervention voltage was exactly $(g_b^E + \kappa_b V_b)/(g_b^L + g_b^E + \kappa_b) = V_b$, while the local eligibility was attenuated by $(g_b^L + g_b^E)/(g_b^L + g_b^E + \kappa_b)$. This state clamp separated physical input-resistance attenuation from forward feature suppression.

Initial full-data gradients defined branch signal energy S_b. Independent coefficient noise was scaled to give $N_b = S_b \exp(-2hx_b)$ for eight equally spaced $x_b \in [-1,1]$ and $h \in \{0, 0.5, 1, 1.5, 2\}$. With the actual step $\eta = c/L$ and $c = 0.5$, shunts were fixed from the initial oracle attenuation $\min\{1, S_b/[c(S_b + N_b)]\}$ or its step-consistent best global, shuffled or reversed control; they were not learned. Other controls were clean exact backpropagation, noisy credit without a shunt, a state-matched additive current, an explicit point gate receiving the identical factors, and aligned attenuation with clean credit. Forty common-noise full-data updates used one half of the numerically estimated initial smoothness limit. This correction was frozen after identifying that the preceding pilot had paired this half step with the $c = 1$ gain. Two new artifact-only canary seeds preceded 50 fresh paired confirmatory seeds 8100–8149. The frozen gates required nonnegative inputs and shunts, voltage matching and independently computed point/additive equivalence within 10^{-12}, and relative finite-difference errors below 10^{-5} for both the unshunted objective and shunted frozen-clamp surrogate. Observed maxima were 2.22×10^{-16} for state mismatch, 4.00×10^{-15} and 1.31×10^{-14} for the equivalence gates, and 3.60×10^{-8} for the physical finite-difference error.

Intervals resampled paired seeds with 20,000 draws; paired tests were two-sided Wilcoxon signed-rank tests. Configuration, runner hashes, all 2,250 seed–condition rows and the prespecified contract are supplied with Source Data.

Same-span noisy coefficient-learning experiment

The rate-based coefficient experiment fixed a 16-coordinate balanced-tree space and the rank-eight span through depth three. Three parameterizations shared this projector: eight orthonormal Haar columns, 15 redundant raw nested indicators and the same indicators multiplied by fixed positive gains spanning $\exp(-3)$ to $\exp(3)$. Their positive Gram condition numbers were 1, 15 and 388.52. Each seed supplied a unit-norm target in the common span and 80 common noisy target estimates. Observation standard deviation was $2/\sqrt{n_{\text{eff}}}$ for $n_{\text{eff}} \in \{4, 16, 64, 256\}$. Vanilla coefficient descent used 0.2 of each dictionary's stability limit. A separate oracle control multiplied the coefficient gradient by the pseudoinverse Gram matrix, producing the same projected field step in every parameterization.

The design and predictions were frozen before canary seeds 9098–9099 and 50 paired confirmatory seeds 9100–9149. All 4,800 checkpoint rows completed. Projectors agreed to 9.44×10^{-16}, target address residual was at most 1.45×10^{-30}, and Gram-preconditioned trajectories agreed to 1.16×10^{-15}. The unconditional orthonormal-coordinate prediction failed in the low-data regime. We therefore report both the prespecified contrast and the exact post-result bias–variance decomposition rather than relabeling the outcome as confirmatory. Intervals resampled paired seeds with 20,000 draws and tests were two-sided Wilcoxon signed-rank tests.

The historical inhibitory-dose cohort used the noise-resilience task and crossed two neuron models, five inhibitory-contact counts per branch (0, 5, 10, 20 and 40), backpropagation or one of the three local feedback fields, and ten paired seeds, for 400 runs. Its prespecified endpoint was a shunting-minus-additive contrast. Because every shunting cell used the invalid signed-input convention, the complete family is retained in the run-validity ledger but excluded from figures and inference.

The replacement exact-transport/backpropagation audit crossed the two tasks, two cores, depths one to four and paired seeds 42–51, giving 320 runs and 160 method pairs. Training used detached tracked-clean source commit `74792ca` and nested synthetic-data commit `4f3612a`. The noise- task additive core retained signed inputs; the positive-conductance shunting core used an explicit ReLU transfer before conductances were formed. Acceptance required resolved configurations and seeds, complete learning summaries, finite held-out metrics, finite stage-complete checkpoints, current scheduler logs without a fatal error or transport-shape fallback, and clean source identity. Exact-minus-backpropagation effects were paired within task, core, depth and seed; the plotted task–core contrasts first averaged four depths within seed. Incomplete setup and resource attempts supplied no endpoint and were excluded independently of outcome.

The fixed-topology cohort contained 320 runs at depth four and crossed the two tasks, neuron models, backpropagation or three local feedback fields, random or spatially partitioned sparse input indices, and ten paired seeds. Both maps had 21 contacts per distal branch. The spatial map recursively divided the 28×28 feature plane into 16 disjoint rectangles per neuron; the random map sampled each branch independently from all 784 coordinates. A configuration-level audit quantified unique coordinate coverage and cross-branch overlap for the same ten topology seeds. The validity rule removed the 80 synthetic-noise shunting runs, leaving 240 comparisons. Because any spatial effect shared by backpropagation is a forward-connectivity effect, this cohort was not used to claim topology-specific credit transport.

The fixed-budget depth cohort used the synthetic task and contained 320 runs. The retained 160 additive runs crossed four depths, three local feedback fields and matched backpropagation at ten paired seeds while fixing 16 terminal branches and 960–968 active contacts per soma. This matches terminal branches and input contacts, not compartment or trainable-parameter count.

For the prospective fixed-checkpoint diagnostic, scalar, neuron-indexed, and exact teaching fields were evaluated at the 120 input-valid backpropagation checkpoints on one frozen test batch of 128 examples. Only non-somatic dendritic stages and their trainable branch parameters entered the field and gradient vectors. Scaled capture was one minus the squared residual after optimally rescaling a candidate vector to the exact vector. For the learning endpoint, each candidate gradient was normalized to the exact-gradient norm and applied without further training. Step size was $r\|\theta\|/\|\nabla\mathscr{L}\|$ for $r \in \{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}\}$; 10^{-5} was the primary plotted operating point and all four are supplied as source data. Relative progress was the candidate loss decrease divided by the loss decrease from the norm-matched exact update. The association interval resampled checkpoints while retaining the paired scalar and neuron-indexed fields.

MICrONS morphology processing

Cells were drawn from the functionally co-registered MICrONS cohort and resolved from stable nucleus identifiers to roots in materialization 1822 of `minnie65_public`. Skeletons were downloaded through the MICrONS skeleton service. Incoming synapses came from `synapses_pni_2`. The original eight cells were a convenience pilot selected under the earlier analysis contract, not a random or representative sample; every anatomical claim from this cohort is therefore explicitly single-animal and descriptive. Degree-two dendritic chains were collapsed into branch-to-branch segments. Synapses were assigned to the nearest dendritic cable node and retained within $5\,\mu$m. Presynaptic coarse type supplied E/I identity where available; otherwise, a high-confidence target-structure prediction supplied a secondary proxy from the in-development `synapse_target_predictions_ssa_v2` table. This table does not currently have a stable public release identifier and is not required for the direct-type-only or disjoint-v661 analyses. The direct-type-only analysis discarded all proxy-classified contacts.

Raw axial coupling was proportional to radius squared divided by cable length; it was normalized by its nonzero cell median and clipped to 0.05–20. Raw leak was proportional to radius times the larger of cable length and radius; it was normalized by its nonzero median, and was clipped to 0.1–10. Excitatory and inhibitory synaptic areas were normalized by the median nonzero total synaptic area within cell. These quantities define sensitivity weights, not fitted physical conductances. The soma was the root of every compressed tree. Ancestry matrices were computed by explicit parent traversal and checked for acyclicity and a unique path to the soma.

For the disjoint sensitivity cohort, we froze a 55-cell eligibility universe before inspecting outcomes, mapped all candidates to the public MICrONS v661 static release, and excluded the original eight cells by stable nucleus identifier. All 47 remaining cells passed the structural criteria. Only direct presynaptic E/I annotations were used. Direct labels were available for 11,024 of 234,301 incoming synapses before spatial mapping (4.71%); 10,516 contacts mapped to retained cables (median 212 directly typed synapses per cell). The cohort was not a population-random sample and contained 34 L2 IT, 2 L3 IT, 10 L4 IT, and 1 L5 ET cell. The same structural-capacity procedure was applied without pooling the two cohorts.

MICrONS inhibitory-census analysis

The inhibitory analysis joined the 47-cell v661 cohort to the published v795 census by stable nucleus identifier, before evaluating any endpoint. All 20 overlapping targets were retained. We used the published cell-type table, inhibitory-synapse table, complete input-synapse table, and target SWCs [33]. Raw all-input coordinates were mapped into the published slanted micrometre frame with an affine transformation fit from 80,114 synapse locations represented in both coordinate systems. The maximum transform residual was $1.64 \times 10^{-12}\,\mu$m. Input and known inhibitory locations were mapped to the nearest target skeleton node and retained within $5\,\mu$m. Inputs without a known presynaptic inhibitory cell are called generic rather than excitatory because the census does not label every incoming contact.

For a multisynaptic inhibitory connection, one largest contact was retained within each published axonal clump. Every retained contact was matched to 64 generic input segments in the same compartment with nearest log path distance to the soma. Five hundred independent matched assignments estimated the null within each connection. Pairwise outcomes were cable-tree distance, normalized shared path to the lowest common ancestor, and cosine overlap between the two input-size-weighted binary descendant domains. Outcomes were first averaged over inhibitory connections within each postsynaptic target. The target, not a contact or connection, was the inferential unit.

Two sensitivity analyses tested dependence and spatial matching. First, connection-level contrasts were averaged within each of 92 presynaptic axons before inference. Second, each candidate location was ranked jointly by its log path-distance and three-dimensional Euclidean proximity to the observed contact, using the sum of the two within-pool percentile ranks; the same 500 matched assignments were then repeated. These analyses ask whether a target-level contrast persists after repeated connections from one axon are clustered and after local tissue proximity is included in the null.

For route capacity, rows were segments with mapped input and columns were segments bearing at least one known inhibitory contact. The binary ancestry matrix was weighted by the square root of mapped input size by row and known inhibitory size by column. Actual sites were ranked by column norm. At each budget, we reconstructed the full weighted actual-route matrix from the selected columns. Controls were 200 random subsets of actual sites, depth bins, depth-preserving within-column ancestry shuffles, and an unconstrained SVD oracle. The leverage selection is a capacity diagnostic, not a biological learning rule. Because the selected columns and reconstruction target come from the same actual-route matrix, this endpoint measures internal dictionary compressibility rather than generalization to an independently generated credit field.

Credit dictionaries and capture

The weighted direct route matrix was $\Gamma_{ik} = A_{ik}\beta_k$. It was a normalized route-sensitivity proxy, not the exact derivative of the reciprocal passive cable solve. Perturbation fields were sampled by multiplying independent route amplitudes by K. At each channel budget, ancestry routes were ranked by β_k times the fraction of total mapped excitatory synaptic area among their descendants. Controls were an unconstrained principal-component basis, random nonempty real paths, bins of cable depth, and column-wise ancestry shuffles. Channel count and the number of evaluated examples were matched. Feedback wiring density was the fraction of nonzero entries in the site-by-channel dictionary.

For each dictionary, weighted least squares supplied the optimal projection after mapping both the field and dictionary into the synaptic-area-weighted coordinates of Eq. 12. Twenty independent Monte Carlo streams, each containing 384 perturbation fields and 64 random or shuffled dictionaries per cell and channel count, measured numerical sensitivity. Streams were not treated as biological replicates. Primary comparisons were averaged within cell before paired testing.

For the independent-generator control, the reconstruction target was not sampled from K. We added a small focal shunt at every inhibitory-bearing segment of the reciprocal passive-cable system, restored the baseline somatic voltage with an independently solved state-matching compensatory current, and finite-differenced the log magnitude of the exact excitatory-conductance gradient at every excitatory-bearing segment. The derivative dose was 10^{-3} times the focal segment's membrane-plus-synaptic conductance. This produced a weighted response operator H whose columns were exact focal responses. Expected field-energy capture for a dictionary Φ was evaluated without Monte Carlo sampling as

$$\frac{\|P_{W^{1/2}\Phi} W^{1/2} H\|_F^2}{\|W^{1/2} H\|_F^2}, \tag{20}$$

where W contained mapped excitatory synaptic area and P denoted the orthogonal projector onto the stated column space. In addition to the primary controls, 200 surrogate trees per cell reassigned child subtrees among parent slots at the preceding depth. This preserved every node and edge attribute, every node's topological depth, and every parent's out-degree while changing ancestral membership. A separate row shuffle preserved the selected route values and sparsity while breaking their assignment to excitatory segments.

Focal conductance perturbations

Before running the focal analysis, we fixed the site-selection rule, primary dose, comparison groups, and localization endpoint. Sites were selected by a depth-stratified rule and required at least three excitatory descendants and three comparison sites; at most 16 sites were sampled per cell across topological-depth quartiles. Primary excitatory and inhibitory synaptic scales were both 0.35, with normalized reversal potentials $E_E = 1$ and $E_I = -0.2$. Conductance doses were 0.25, 0.5, 1, and 2 times the local membrane-plus-synaptic conductance. The additive control injected the first-order current $\kappa_k(E_I - V_k)$ at the same node without changing G. An independently solved state-matching compensatory current injection restored baseline somatic voltage after either intervention. This is not the derivative of a continuously adjusted voltage-clamp controller.

The objective was squared somatic error, with the target voltage set one normalized unit below baseline. Excitatory-conductance gradients were obtained by an adjoint solve, treating the solved compensatory current as fixed during the post-intervention derivative, and were checked by finite differences. If $d_i = E_E - V_i$ and $d_i' = E_E - V_i'$, the post-shunt gradient $q_i' d_i'$ was decomposed algebraically into an adjoint-only substitution $q_i' d_i$ and a driving-force-only substitution $q_i d_i'$, alongside the full-shunt and matched-additive gradients. Two-factor Shapley values allocated the nonlinear full-shunt-minus-additive localization payoff between the adjoint and driving-force changes and reconciled exactly to the observed contrast at each site. The matched-additive reference is not a literal no-factor physical intervention, so an unperturbed-reference allocation was also computed as a sensitivity analysis. These substitutions are not independent biological interventions. Synapses were classified as descendants, ancestors, sister-subtree sites, or unrelated sites. Unrelated synapses within each template were matched to descendant cable depth. For baseline gradient g_i and perturbed gradient g_i', we defined $m_i = |\log[(|g_i'| + \varepsilon)/(|g_i| + \varepsilon)]|$, where $\varepsilon = 10^{-15} \max(1, \max_i |g_i|)$. The frozen site-level localization index was the median m_i among descendants minus the median among depth-matched unrelated synapses. Site effects were averaged within cell; the primary paired contrast used the eight cell means. A topology control sampled 200 relation templates from other focal sites, stratified by focal-site topological-depth quartile when another site was available; otherwise it used another site in the same cell. Axial and leak normalization and clipping are specified in Supplementary Section S6.

We separately calibrated the passive cable in physical units. Non-root segments were treated as cylinders with axial conductance $\pi r^2/(R_a \ell)$ and leak conductance $2\pi r \ell / R_m$; the root used the spherical membrane area $4\pi r^2$. Conductances were expressed in nS. Synaptic areas are not conductance measurements, so excitatory and inhibitory synaptic conductances retained the prespecified relative scale and were referenced to the cell-median physical leak conductance. We evaluated $R_a \in \{100, 150, 250\}\,\Omega\,\mathrm{cm}$ at $R_m = 15{,}000\,\Omega\,\mathrm{cm}^2$ and, at $R_a = 150\,\Omega\,\mathrm{cm}$, $R_m \in \{300, 1000, 3000, 5000, 15{,}000, 30{,}000\}\,\Omega\,\mathrm{cm}^2$. The same sites, state-matching current injection, adjoint gradient, unit relative dose, and additive comparison were used. Two zero-length non-root segments in the v661 source skeletons used their radius as the cable length; this fallback and all electrotonic ratios are recorded in source data.

The frozen passive focal-sensitivity matrix used the original eight cells at $R_a = 150\,\Omega\,\mathrm{cm}$. We crossed membrane resistivities of 300, 1000 and $15{,}000\,\Omega\,\mathrm{cm}^2$ with distributed leak-like background conductance equal to 0, 1 or 4 times each segment's baseline leak. At every selected site we evaluated three dose schemes: 0.25, 1 and 2 times local membrane-plus-synaptic conductance; fixed doses 0.05, 0.5 and 5 nS; and 0.25, 1 and 4 times local input conductance $[(G^{-1})_{kk}]^{-1}$. Rank-one voltage and adjoint updates were computed exactly with Sherman–Morrison, followed by the same state-matching somatic current. In addition to absolute localization, the frozen outputs included signed log-magnitude change, attenuated and enhanced fractions, sign-flip fraction, descendant gradient-energy ratio, local input resistance and S_k from Eq. 14. Site values were averaged within cell before eight-cell intervals and shunt-minus-additive contrasts. A numerical canary required positive-definite matrices and somatic residual below 10^{-10} before the full matrix ran with unchanged code and configuration.

For the active steady-state sensitivity, we added voltage-dependent Na, K, Ca, HCN and NMDA currents to the physical passive system at $R_a = 150\,\Omega\,\mathrm{cm}$, $R_m = 1{,}000\,\Omega\,\mathrm{cm}^2$ and one baseline-leak unit of distributed background conductance. Channel activation followed sigmoidal steady-state gates with channel-specific half-activation, slope, power and reversal values fixed in the machine-readable protocol. Global and segment-level maximal-conductance multipliers were independent log-normal draws. Newton iterations with residual-reducing line search solved the full nonlinear equilibrium. Draws were accepted only when the voltage remained within the prespecified range, the residual was below 10^{-10} and the symmetrized local Jacobian was positive definite above 10^{-8}. With purely local voltage-dependent gates, the off-diagonal Jacobian entries are the reciprocal axial couplings, so the steady-state Jacobian is symmetric and the symmetrized gate and the plain inverse used below are exact rather than approximations.

For every accepted equilibrium, the adjoint used the exact inverse of that local Jacobian. Focal rank-one shunts and first-order-current-matched additive inputs were evaluated at 0.25, 1 and 4 times local input conductance, followed by the same state-matching compensatory somatic current. We required 64 accepted draws in every cell, up to 256 attempts; all 512 first attempts were accepted. The 64 channel draws and up to 16 focal sites were averaged within each reconstructed cell before bootstrap intervals and paired tests. The active ensemble is a steady-state local-linearization sensitivity, not a simulation of channel kinetics or spikes.

The focal analysis was also repeated after removing all proxy-classified contacts. In the disjoint v661 sensitivity cohort, 45 of 47 cells supplied 235 eligible focal sites for the primary shunt–additive comparison. Forty cells with at least two eligible sites supplied the reassigned-relation control. Cells, rather than focal sites, remained the units of inference.

Measured-response analyses

Functionally imaged presynaptic partners were linked to their anatomical contacts through the MICrONS/CAVE and DANDI mappings. The original frozen one-scan cohort included 69 partners across seven targets. The scan-complete analysis deterministically retained every automatic-conservative target–session–scan cohort with at least five unique connected imaged presynaptic roots, giving 13 scan observations nested within the same seven targets. Each scan contained 464 trials, 280 stimulus identities and 136 repeated identities. Multiple contacts from one presynaptic partner were pooled; its analysis site was the segment with greatest summed contact area. Functional similarity was the Pearson correlation of condition-mean responses over repeated identities. For each target, the partial-rank analysis rank-transformed similarity and shared ancestry, separately residualized both against log Euclidean separation and path-depth difference, and correlated the residuals. Scan-level statistics were averaged within target before the target bootstrap and signed-rank test. The target neuron, not a scan or synaptic pair, was the inferential unit.

For task-derived credit, a small branch model predicted each postsynaptic response from its connected presynaptic responses. Complete visual-stimulus identities were assigned to held-out splits. Exact branch errors were computed on training trials, and dictionary projectors were fit without held-out identities. Matched models were then trained from the same initialization with exact or projected branch errors. Ten fixed split/initialization replicates were averaged within each of seven target cells. Because projection coefficients used exact errors, these analyses estimate the capacity of a feedback family rather than a biological encoder. Only input-weight errors were projected; branch-readout weights and bias received exact gradients. Normalized MSE was held-out MSE divided by the held-out error of the training-mean predictor. Complete model and optimizer specifications are in Supplementary Section S9.

For the segment-resolved analysis, each complete compressed reconstruction was instantiated as a reciprocal passive conductance system. A trainable non-negative excitatory conductance was placed at the dominant mapped contact of every imaged partner. Exact gradients of somatic squared prediction error used the trial-specific inverse conductance operator and factored into $x_i(E_E - V_i)$ times the segment adjoint. Restricted methods first projected the baseline soma-to-site transfer vector into a four-channel conductance-weighted ancestry dictionary, a column-value-preserving site shuffle or an equal-rank random set of nonempty anatomical routes. This vector was calibrated once without target responses and frozen during learning; each restricted update then used only scalar output error, the fixed site vector and local eligibility. Exact readout and bias gradients were used in all methods. Ten seed-fixed splits held out complete stimulus identities within every scan. Replicates were averaged within scan and all eligible scans within target before seven-cell inference. Finite differences audited one exact conductance coordinate per run, input hashes and runner hash were archived, and any absent route, rank mismatch, non-finite value or incomplete scan caused exclusion; none occurred.

The capture–learning association excluded the exact and dense oracles. Both variables were centered within target before computing Spearman's ρ. Its interval used 20,000 target-level bootstrap draws, and its two-sided null used 20,000 independent reassignments of learning outcomes among method labels within each target.

Alignment-controlled optimization

For each reconstructed cell, we rebuilt the conductance-weighted ancestry kernel and selected eight routes by inhibitory fraction times descendant excitatory synaptic-area fraction, before generating gradient fields. Segment coordinates were weighted by the square root of summed mapped excitatory synaptic area. If Q is an orthonormal basis for the selected routes, each unit-energy exact gradient was constructed as

$$\boldsymbol{g}(a) = \sqrt{a}\,\boldsymbol{u}_{\parallel} + \sqrt{1-a}\,\boldsymbol{u}_{\perp}, \tag{21}$$

where $\boldsymbol{u}_{\parallel} \in \mathrm{span}(Q)$, $\boldsymbol{u}_{\perp} \perp \mathrm{span}(Q)$, and $a \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$. Directions were paired across alignment levels. Eight random real paths, eight cable-depth bins, and row-permuted ancestry routes supplied controls. The permutation preserved column values and nonzero count while breaking ancestry.

Each cell contributed 40 nested Monte Carlo streams with 256 gradients per stream. The objective was a positive diagonal quadratic with an anatomy-independent curvature spectrum from 0.5 to 1.5, permuted within stream and fixed across methods and alignment levels. One-step updates had norm 0.1 times the exact-gradient norm. Iterative projected descent used learning rate 0.25 for 20 steps. Both outcomes were normalized by the corresponding full-gradient progress. Streams and gradient fields measured numerical sensitivity; cell-level means were the inferential units.

External animal-data reanalysis

Francioni et al. source data were obtained from the public Nature record associated with DOI 10.1038/s41586-026-10190-7. The archived workbook was verified by SHA-256 before analysis. Extended Data sheet 13E supplied one P+ and one P- contrast per animal, defined by the source study as the change in soma–dendrite residual between error-reduction and error-increase periods. We retained the source signs. Signed separation was P+ minus P- and was checked against the value reported in the workbook. The common basis was $(1,1)/\sqrt{2}$ and the signed basis was $(1,-1)/\sqrt{2}$; squared projection norms were divided by total two-coordinate energy. Sheet 5E supplied neuron-level distributions for visualization only. Directional animal-level hypotheses followed the known causal mapping. We report exact random-sign and Wilcoxon tests together with 50,000-draw animal bootstrap intervals.

Static task-interference calculation

The task-interference identity follows from exact quadratic expansion around a Task-A optimum, where the linear term vanishes. For its equal-route illustration, each task occupied 1,000 coordinates, route overlap varied from zero to one, and the Task-B target opposed Task A on shared coordinates. Off-route leakage supplied an update of relative magnitude λ on Task-A-only coordinates. We evaluated a 101×101 grid over overlap and leakage at step size 0.2. Explicit vector updates were compared with the closed form at every grid point. This analysis contains no temporal or spiking neural dynamics and uses the published motor-learning study only for qualitative, cross-study interpretation.

Interior-optimum and wiring-normalized reanalyses

Two deterministic secondary analyses of frozen outputs support Supplementary Fig. S16 and Fig. 6k,l; neither alters or reruns any training outcome. For the interior-optimum analysis, the per-seed initialization utility of Eq. 9 was taken from the archived operator metrics of the 2,700-fit factorial and from the frozen depth-phase tables. The predicted optimum is the argmax of that utility over routed depth, or of the ancestry-minus-best-control utility contrast over budget; the observed optimum is the argmin of final population loss, or the argmax of the corresponding held-out accuracy contrast. Before any new quantity was reported, the pipeline reproduced the published $K = 4$ ancestry-minus-best-control accuracy contrast (0.0127; 15 of 20 seeds) from the frozen seed outcomes. For the wiring-normalized analysis, capture per unit wiring is weighted field-energy capture divided by feedback wiring density, computed per cell from the frozen routing-capacity curves after averaging Monte Carlo streams within cell; the published eight-channel capture and wiring anchors were reproduced before any ratio was formed. Intervals are 95% seed- or cell-bootstrap intervals with 20,000 resamples. The laminar-capture proposition and the marginal utility condition (Supplementary Section S4) are verified numerically in the released test suite.

Statistics and reproducibility

All tests were two sided unless a directional hypothesis had been frozen in advance. Effect sizes and 95% intervals accompany central comparisons. Reconstructed-cell intervals resampled cells first and, where appropriate, a Monte Carlo stream within cell. Textual focal-attribution intervals used a hierarchical bootstrap over cells and sites; Fig. 7h instead shows cell-bootstrap intervals after averaging sites within each cell. Measured-response intervals resampled target cells. Artificial-network means and standard deviations used independent initialization seeds. Exact Wilcoxon signed-rank tests were reported for small paired cell cohorts; no site, trial, epoch, or Monte Carlo stream was counted as an independent biological replicate. Null results are described as a lack of reliable evidence rather than proof of equality. Bootstrap intervals used 20,000 draws. The four-budget primary subtree scan used Benjamini–Hochberg correction; other parameter sweeps and sensitivity analyses are labeled secondary and are not treated as independent confirmation.

Seed counts followed the evidential role of each cohort rather than one pooled power calculation: three to five seeds were reserved for numerical or mechanism diagnostics; ten for fresh paired confirmatory replications with interval and sign-agreement gates; 15–20 for inherited trained factorials; and 50 for phase boundaries and crossover distributions. Effect sizes were not known before the new simulation families, so no prospective parametric power calculation was used. Positive confirmatory claims required a paired 95% interval excluding zero and the prespecified fraction of concordant seed signs; diagnostic cohorts were not promoted to inferential evidence. A machine-readable table in Source Data maps every headline endpoint to its independent unit and identifies nested or descriptive observations.

Analysis code records configuration files, seeds, software versions, source tables, and machine-readable summaries. Numerical source data are organized by figure and panel in the accompanying package. Tests cover tree construction, projection identities, and exact-gradient checks.

Reporting of computational assistance

OpenAI Codex was used to assist code development, statistical and consistency audits, figure preparation, literature checks, and language editing. It was not listed as an author and did not determine the scientific conclusions. The authors retain responsibility for the study design, interpretation, accuracy of the code and numerical results, citations, and final manuscript.

Ethics and data reuse

No new experiments involving animals or human participants were performed. All biological analyses reused public MICrONS, DANDI, and published animal source data; experimental collection and institutional approvals are described in the source studies [33, 37, 56].

Data availability

MICrONS anatomy and connectivity are available through the public CAVE datastack `minnie65_public`. The primary cohort used materialization 1822, skeleton-service reconstructions, and the `synapses_pni_2` table. The disjoint sensitivity cohort used the public v661 static release. The secondary target-structure labels used only in the full-contact analysis came from the in-development `synapse_target_predictions_ssa_v2` table and are therefore distributed only as derived labels subject to the source-data access terms; all principal structural findings were repeated without those labels. Volumetric imagery is archived in BossDB under DOI 10.60533/BOSS-2021-T0SY, and co-registered functional data are available as DANDI:000402; exact asset and table identifiers are recorded in the supplied manifest. The source publication describes the full data release [56]. Public benchmark datasets are available from their original repositories. The inhibitory-census tables and SWCs are available with the source record for Schneider-Mizell et al. [33]. The Francioni et al. workbook is available from the Source Data link associated with the Nature article [37]; its checksum and used sheet names are recorded in the manifest. Derived cell and exclusion manifests, numerical source data underlying every figure, and non-restricted summaries are supplied with this manuscript. Authentication tokens and externally hosted raw caches are not redistributed.

Code availability

The accompanying reviewer software archive contains the complete training implementation used here, exact-gradient and perturbation analyses, frozen configurations, tests, figure scripts and byte-identified snapshots of the critical training components under the MIT License. A separate Source Data archive contains the numerical data underlying every display item. Submission is gated on replacing this sentence with the public repository URL and versioned archival DOI; no public identifier is asserted before that record is created.

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Author contributions

H.S. developed the computational framework, performed the analyses, and wrote the initial manuscript. M.R. contributed to the local-credit-assignment theory, implementation, and interpretation. B.L.S. supervised the work and contributed to the biological framing, study design, and manuscript. All authors reviewed and revised the manuscript.

Competing interests

The authors declare no competing interests.